Magnetism in Solids: Hysteresis

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1 Introduction

Magnetic phenomena associated with solids have been known for over two millennia; for example, lodestone (which is better known as magnetite, Fe_3O_4) appeared in Greek and Chinese writings many centuries before the Christian era. Though its physical origin was unknown at the time, magnetism and magnetic materials continued to attract much attention due to their "mysterious" physical properties. The origin of the name *magnet* derives from "the stone of Magnesia", a Greek town and province in Asia Minor. One of the first true scientific works dedicated to magnetism appeared in 1600 with the publication of *De Magnete* by William Gilbert. Research in magnetics in the seventeenth and eighteenth centuries was dominated by the military and in particular by the British Navy. While much progress was made during this time, it was with the developments concerning the unification of electricity and magnetism in the 19th century that enormous strides were made. In 1820, Oersted discovered that an electric current could generate a magnetic field. This was followed by the production of the first electromagnet in 1825 by Sturgeon.

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Warburg produced the first measurement of the hysteresis loop for iron in 1880. While Faraday, Gauss and Maxwell etc. made much theoretical progress in the mid to late 1800s, however, it was not until the 20th century that the foundations were laid for a more comprehensive physical description of magnetic materials. Curie and Weiss developed early models of the phenomenon of spontaneous magnetisation and its variation with temperature. Indeed the critical temperature for the transition from ordered (ferromagnetic) state to the disordered (paramagnetic) state is known as the Curie temperature, T_C. Pierre Weiss postulated the existence of magnetic domains in order to explain the magnetisation of a magnetic material from its virgin state with a zero net magnetisation. The regions between these domains of spontaneous magnetisation are referred to as domain walls and were described in detail by Néel, Bloch and Landau.

Of the magnetic properties associated with magnetic materials, magnetic hysteresis is especially important. Indeed, it is this property which allows ferromagnetic (and ferrimagnetic) materials to retain a magnetised state, or remanence, in the absence of an applied magnetic field. This property alone permits the many applications of magnetic materials, such as the magnetic compass, electric motors and magnetic recording. Magnetic hysteresis is most commonly visualized with the magnetisation – applied magnetic field curve, as illustrated in **Fig. 1**. In this figure we illustrate the main features of the curve, which schematically traces the magnetisation, M, as a function of the applied magnetic field, $H_{applied}$; the saturation magnetisation, M_S; the remnant magnetisation (at $H_{applied}=0$), M_R; the saturation field (the applied field for which the magnetisation is at saturation, $M=M_S$); the coercive field, H_C (which is the applied magnetic field necessary to reduce the magnetisation state to zero, $M=0$). In this representation we assume that the sample is taken to its saturated state, i.e., the largest value of the magnetisation for the sample. This is referred to as a major loop. Minor loops occur when the magnetic field is reduced before the sample is taken to the saturated state, or when the cycle is made between limits that do not include both magnetic saturation points.

Of the magnetic properties encountered in condensed matter physics, diamagnetism and paramagnetism are the most common. In fact, a vast majority of elements and compounds fall into the categories of paramagnetism and diamagnetism, which display weak magnetic susceptibilities, as determined from the relationship between the magnetisation, M, induced by an applied magnetic field, H:

$$M = \chi H \tag{1}$$

For weak applied fields this relation is a good approximation to a linear isotropic homogeneous response for paramagnetic and diamagnetic materials. In the former the magnetic susceptibility is positive and weak, while for diamagnetic systems it is negative and weak. A more sophisticated model for paramagnetic materials was postulated by Langevin in 1905, who developed a semi-classical model for a system of non-interacting spins. Here the magnetisation follows the so-called Langevin function, which can be expressed by:

$$L(\alpha) = coth\ \alpha - \frac{1}{\alpha} \tag{2}$$

with

$$\alpha = \frac{\mu B}{k_B T}$$

where μ is the magnetic moment of a single ion, B is the applied magnetic induction (The relationship between the magnetic induction, B, magnetic field strength, H, and the magnetisation, M, is expressed as: $B=\mu_0(H+M)$), T the temperature and k_B the Boltzmann constant. The magnetisation follows the relation:

$$M = M_S L(\alpha) \tag{3}$$

in which the saturation magnetisation is given by $M_S=n\mu$, where n is the total number of ions per unit volume in the system.

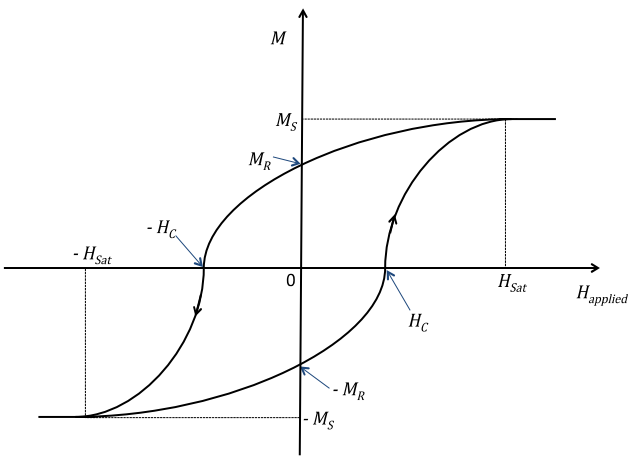

Fig. 1 Schematic illustration of a magnetic hysteresis loop indicating the magnetisation as a function of the applied static external field.

The quantum mechanical approach to this problem allows for the quantisation of the spin of the ions and leads to an expression in which the Langevin function is replaced by the Brillouin function, which takes the form:

$$B_J(\gamma) = \frac{2J+1}{2J} coth\left[\frac{(2J+1)\gamma}{2J}\right] - \frac{1}{2J} coth\left(\frac{\gamma}{2J}\right) \tag{4}$$

with $\gamma = g\mu_B JB/k_B T$, where μ_B is the Bohr magneton, J is the spin number, which defines the quantisation of the spin. It is noted that for a non-quantised, classical system $J=\infty$ and the Brillouin function reduces to the Langevin function.

One of the properties observed from both the classical and quantum models for paramagnetism is that the magnetisation reduces to zero for no applied field and there will be no hysteretic behaviour in the system. Indeed for non-interacting spin systems there will be no hysteresis. It is only when an interaction between spins exists that any cooperative effects are manifest, i.e., via some form of exchange interaction, and hysteretic effects become observable. There are many forms of interaction between spins, the nature of which depends on the material in question. For example, in metallic systems, such as Fe and Ni, direct exchange coupling between neighbouring spins arises from an overlap of electronic wave functions. However, more complex forms of exchange occur in oxide systems in which the coupling between spins occurs via intervening oxygen atoms typically arising in antiparallel alignments. This form of indirect coupling is referred to as super-exchange and is common in antiferromagnetic and ferrimagnetic systems. Only the latter exhibits a spontaneously measurable magnetisation and magnetic hysteresis. Antiferromagnetic systems, such as CoO and MnFe, while being fully magnetically ordered, have no net magnetisation, since oppositely aligned magnetic sublattices are fully compensated. In ferrimagnetic systems, the differing populations of oppositely aligned magnetic sublattices mean that a net moment per unit cell is non-zero, or for spins originating from different atomic/ionic species for the different sublattices, which do not fully cancel. The measured magnetic properties of these materials resemble those of ferromagnetic systems, including spontaneous magnetisation, magnetic domains and of course hysteretic behaviour. Temperature dependent magnetisation can be rather different however due to differing temperature dependencies of the magnetic sublattices, which can also give rise to a compensation temperature at which the global magnetisation disappears. The ordering temperature in these materials is known as the Néel temperature, T_N, in honour of Néel's work on the development of the theory of ferrites.

Many applications of magnetic materials rely directly on the hysteretic behaviour for their specific functions. For example, magnetic recording media require a reasonably strong coercive field to provide sufficient stability to store information on magnetic bits, however this should not be too elevated since the writing process would become problematic. In this short contribution we aim to provide the reader with a basic introduction to the phenomenon of magnetic hysteresis where we outline the physical origins and a theoretical description. We also give an outline of the principal experimental methods used to characterise magnetic hysteresis in solids. Finally we will discuss hard and soft magnetic materials and some applications of magnetic materials that rely on specific hysteretic behaviour.

2 Theory of Magnetic Hysteresis

There are many models that have been developed over the past century which explain the phenomenon of magnetic hysteresis. In the following sections we will outline some of the principal elements of the most common of these models, which will illustrate how hysteresis can arise from the fundamental properties of magnetic materials.

2.1 Models of Hysteresis

Historically there have been many attempts to explain the phenomenon of hysteresis and in general the variation of the magnetisation of a ferromagnetic material as a function of the applied magnetic field. In early approaches the explanations can broadly be categorized into two main groups; the first was based on a frictional type force, while the second considered the strong mutual interactions between the magnetic moments of individual atoms. The original idea for the existence of a restoring force to maintain moments in their initial unsaturated state was postulated by Weber (1852), which while possibly providing a mechanism for the initial magnetisation curve would not explain the remnant state after the removal of a magnetic field. Wiedemann (1886) considered the frictional resistance to rotation of magnetic moments to account for the effects of hysteresis. It was Maxwell (1873) who appears to have been the first to consider the effects of mutual interactions between arrays of magnetic moments to explain the effects of hysteresis. This was further developed by Ewing (1890) who believed that such interactions between magnetic moments could fully account for the observed magnetic hysteresis in magnetic materials.

As will be seen in the following section, strong interactions between magnetic moments, which maintain them in a ferromagnetic state, can indeed give rise to hysteretic behaviour. However, magnetic reversal can be rather abrupt and is strongly anisotropic, a fact not incorporated into the early attempts at an explanation for hysteresis phenomena. In the frictional force based models, smoother changes to the magnetisation were envisaged and can be related to the effect of pinning of domain walls by defect sites within the solid. This produces a force opposing the changes to the magnetisation that would otherwise occur under the influence of an applied magnetic field.

Progress in the modelling of hysteresis behaviour was hampered by their restriction to specific ranges of applicability. For example, Cullity (1972) finds only three cases where algebraic expressions were obtained for magnetisation curves. These are for high field magnetisation curves in single crystals, Williams (1937), in polycrystalline materials, where the law of approach to

saturation governs the variation of the magnetisation, Chikazumi (1964) and finally for low-field magnetisation curves and hysteresis loops for polycrystals that display Rayleigh loops, Rayleigh (1887).

In the following we review some of the principal models that have been developed for a more in-depth critique of the basic principles of the approach to the simulation and modelling of hysteresis phenomena. In each case there exist variants and modified versions which build on the pioneering ideas of each model. We will mainly consider these basic principles.

2.1.1 Stoner – Wohlfarth model

The Stoner – Wohlfarth (SW) model, which dates back to 1948, is a simple attempt to evaluate the hysteretic behaviour of ferromagnetic systems (Stoner and Wohlfarth, 1948). It has subsequently become a very important first approximation method for the study of ferromagnetic nanoparticle systems. In the SW model it is assumed that the particle or magnetic body has a single magnetic domain with uniaxial anisotropy. This assumption is based on the original premise of the model that the magnetisation remains constant throughout the sample. This has the consequence of maintaining the total exchange energy constant during the magnetisation reversal process. Such behaviour is termed uniform or coherent rotation and is also appropriate for weakly or non-interacting magnetic nanoparticle assemblies. The model breaks down for very small nanoparticles, where the surface anisotropy is different from the bulk and non-collinear spin configurations are predominant.

The magnetic anisotropy of the particle, which we will consider as uniaxial, can be of magnetocrystalline or magnetostatic origin, since it has no important bearing on the outcome of the model. We use an effective anisotropy for this reason. As such, we can express the free energy density of the system as:

$$E_{SW} = K_{eff} sin^2\theta - \mu_0 M_S H cos(\theta - \phi) \tag{5}$$

where angles are defined in **Fig. 2**. For our definition we use $K_{eff} > 0$, so that an energy minimum occurs for $\theta = 0$, i.e., with the magnetisation along the easy axis. The effective anisotropy can, for a single small magnetic nanoparticle, be expressed as (Skomski, 2008; Tannous and Gieraltowski, 2008): $K_{eff} = K_1 + \mu_0 (N_\perp - N_\parallel) M_S^2/2$. The constant K_1 refers to the uniaxial magnetocrystalline anisotropy constant which aligns the individual atomic magnetic moments along its easy-axis, the second term originates from the magnetostatic anisotropy, due to the shape of the particle, where $N_\perp$ and $N_\parallel$ are, respectively, the demagnetising factors perpendicular and parallel to the easy-axis. For this example we consider the case where these contributions are aligned to minimise the energy density along the same axis, as illustrated in **Fig. 2**. For further discussion of such considerations, see Skomski (2008). Other configurations are possible as is the possibility of other forms of magnetocrystalline anisotropy. The example we give here is the simplest model possible and allows us to demonstrate the appearance of magnetic hysteresis.

In **Fig. 3** we show the energy landscapes for some applied magnetic field values. The energy minima denote the orientations for the stable configuration of the magnetisation vector. For zero and low applied (reversal) magnetic fields, there are two stable minima at $\theta = 0$ and $\theta = \pm\pi$. As the reversal field increases ($H < H_K$ where $H_K = 2K_{eff}/\mu_0 M_S$ is called the anisotropy field), one of the minima will be deeper than the other; this defines the most stable configuration of the system. We note that the energy barrier between the global minimum and the local minimum (meta-stable state) reduces. At an applied field of H_K the local minima vanishes, this corresponds to the coercive field of the system. If the magnetisation happens to be "trapped" in this local minima then at the point where $H = H_K$, it will switch or reverse its direction. Clearly, depending on the sample history, i.e., the state of the sample when we start applying the magnetic field, the magnetisation remains in its original state or will reverse. This signifies the origin of hysteresis in its simplest form and arises due to the fact that the spins act as a macrospin state and the magnetic anisotropy produces a coercive field. This is a consequence of the exchange interaction between spins, which is directional and leads to magnetocrystalline anisotropy in solids.

The equilibrium orientation of the system is defined by the configuration which gives a minimum energy, indicated as $\theta = \theta^*$. The condition for minimising the free energy is given by:

$$\left(\frac{\partial E_{SW}}{\partial \theta}\right)_{\theta = \theta} = 0 \text{ and } \left(\frac{\partial^2 E_{SW}}{\partial \theta^2}\right)_{\theta = \theta} > 0 \tag{6}$$

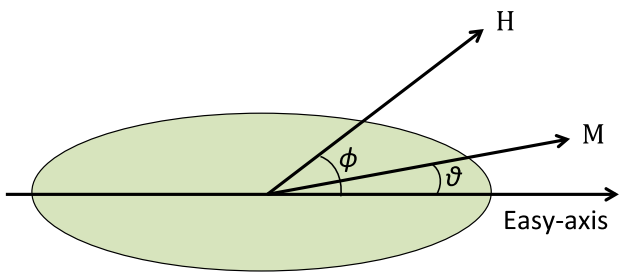

Fig. 2 Schematic diagram for a magnetic single domain with uniaxial anisotropy. Relative orientations of the applied magnetic field, *H*, and the magnetisation, *M*, are shown with respect to the easy axis.

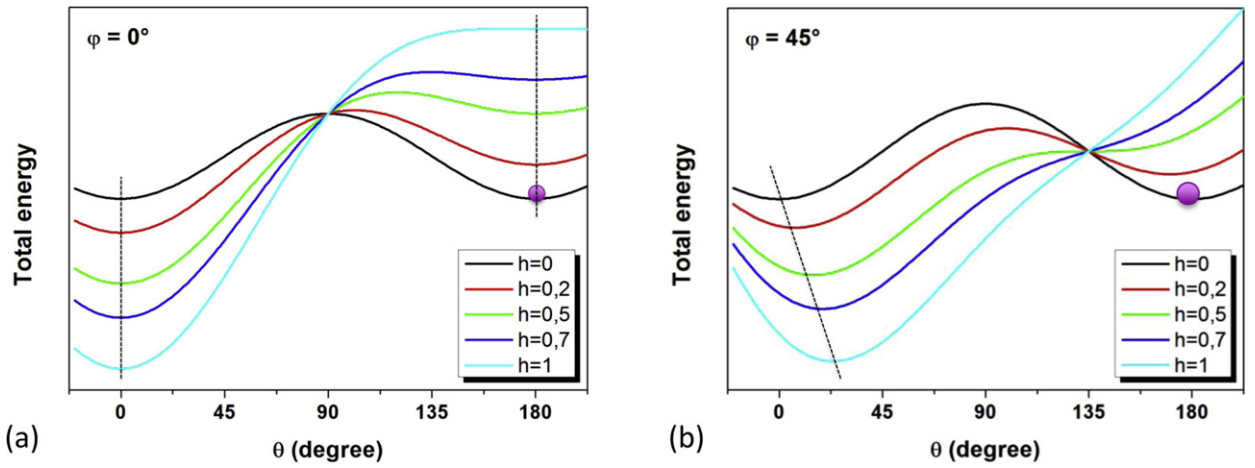

Fig. 3 Energy landscape for the case of the (a) aligned and (b) non-aligned Stoner–Wohlfarth model. In the aligned case, the landscape exhibits two minima at low fields, one at $\theta=0$ (↑) and the other at $\theta=180^{\circ}$(↓). As the field increases and approaches the coercive field, the local minima become very weak, and finally disappear for $H>H_K$ and only the (↑) minima remain. In this case the position of the minima does not change. For the case where the applied field is not aligned with the easy axis, as the field increases the positions of the energy minima change and tend towards the direction of the applied field. As H approaches H_K, the shallow local minimum gradually disappears and the system will stabilize in the deeper energy minimum.

From this we obtain:

$$[sin\theta cos\theta - hsin(\theta - \phi)]_{\theta = \theta^*} = 0 \tag{7}$$

and

$$[cos2\theta + hcos(\theta - \phi)]_{\theta = \theta^*} \geq 0 \tag{8}$$

where we have used the following reduced quantities: $h=H/H_K$ and $m=M/M_S$. The calculations of the magnetisation state are made by varying the strength and direction of the applied magnetic field, where the magnetisation will respond by aligning in that field due to the relative strength and orientation of the magnetic anisotropy as well as that of any applied magnetic field. It is customary to decompose the magnetisation into parallel and perpendicular components, which we express as: $m_{\parallel}=cos(\theta-\varphi)$ and $m_{\perp}=sin(\theta-\varphi)$. The variation of these transverse and longitudinal components of the magnetisation is illustrated in **Fig. 4** for different orientations of the applied magnetic field.

In the Stoner–Wohlfarth model the hysteretic magnetic behaviour arises from the magnetic anisotropy, where the coercive field, H_C, is related to the anisotropy field. The model is also used to show the magnetisation reversal or switching behaviour in single domain particle systems. This is achieved by evaluating the boundaries of the hysteretic behaviour via the first and second derivatives of the free energy. In fact the entire model is based on the variation of the energy landscape, in which the minima determine the orientation of the magnetisation, which is affected by the strength and orientation of an applied magnetic field with respect to those of the magnetic anisotropies, as illustrated in **Fig. 3**. The switching field dependence can be expressed in terms of the anisotropy and orientation of the applied magnetic field in the form:

$$H_{SW} = \frac{H_K}{(sin^{2/3}\phi + cos^{2/3}\phi)^{3/2}} \tag{9}$$

This is displayed for the applied fields in the parallel and perpendicular directions. The form of the switching field curve is commonly known as the Stoner–Wohlfarth astroid. This is illustrated in **Fig. 5**, where we show the theoretical switching field curve.

While the Stoner–Wohlfarth model is a good first approximation, it does assume that the magnetisation is always homogeneous. This situation is generally favoured only in small magnets, which have the form of an ellipsoid of rotation. This means that the particle acts as a single macrospin and the exchange energy is a dominant energy in the system. Deviations from this model can occur for a number of reasons, such as edge and surface anisotropies, which can cause some form of non-collinear alignment of the magnetisation at the surface, meaning $\nabla M=0$. Defects and other magnetic inhomogeneities will also bring about deviations from the model and add further contributions to the coercive field, H_C, which in the Stoner–Wohlfarth model arise only from the anisotropy. Non-coherent reversal of the magnetisation can also occur in elongated single domain particles and therefore is not well described by the SW model. Such processes are usually described by a curling of the magnetic moments in the particle.

2.1.2 The Preisach model

The Preisach model of hysteresis was an early theoretical model of hysteresis dating from 1935. This model can be viewed as a phenomenological representation of hysteresis and has a number of variants that have been developed to overcome some of the

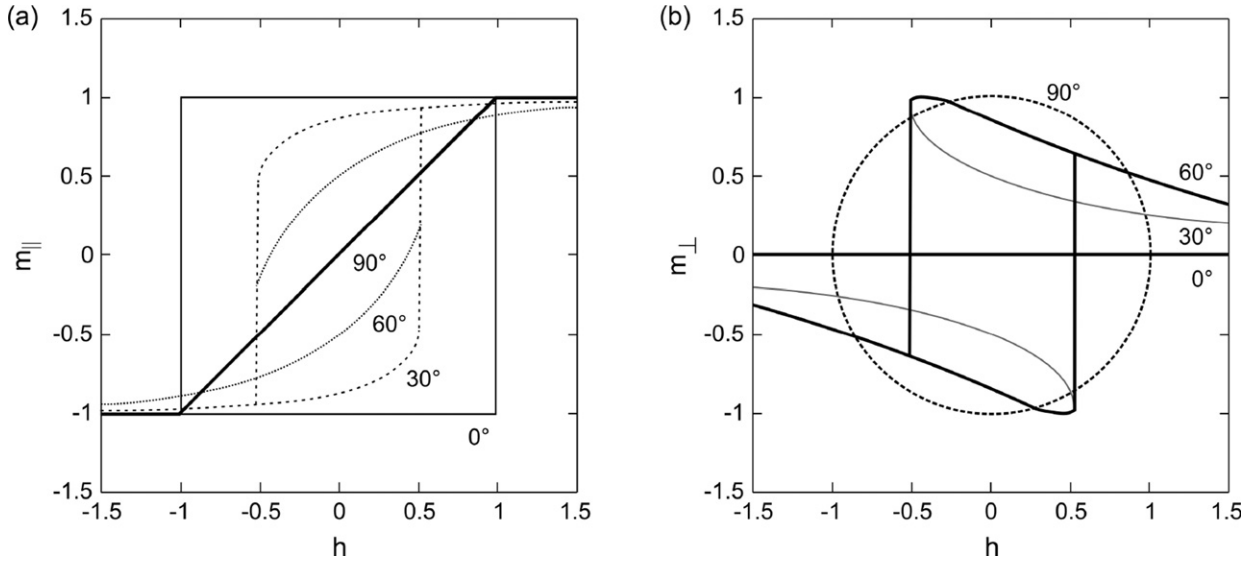

Fig. 4 (a) Longitudinal, $m_{\parallel}$, and (b) transverse, $m_{\perp}$, components of the magnetisation. The angles refer to the direction of the applied magnetic field for $\phi = 0°$, 30°, 60° and 90°. Reproduced from Tannous, C., Gieraltowski, J., 2008. "The Stoner – Wohlfarth model of ferromagnetism". Eur. J. Phys. 29, 475–487.

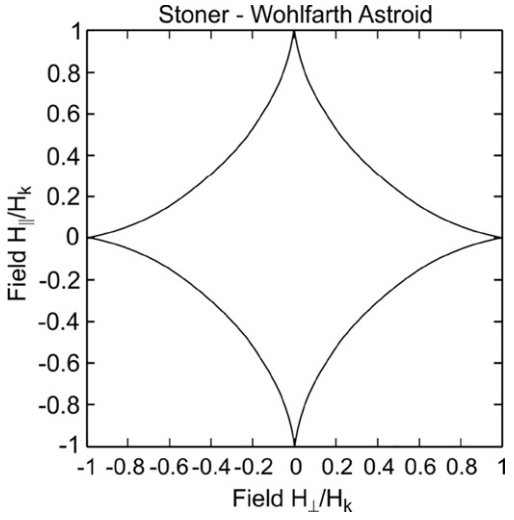

Fig. 5 Switching fields for a single domain particle with uniaxial anisotropy. Due to the shape of the curve, this is popularly known as the Stoner–Wohlfarth astroid. Inside the astroid, the particle is capable of switching, while outside it is not. Reproduced from Tannous, C., Gieraltowski, J., 2008. "The Stoner – Wohlfarth model of ferromagnetism". Eur. J. Phys. 29, 475–487.

shortcomings of the earlier models. In the following we shall present only the basics. Essentially the Preisach model uses the hypothesis that the free-energy profile of a magnetic system is characterised by multiple minima and meta-stable states that can be decomposed into a set of many elementary bi-stable contributions (Benabou *et al.*, 2003). The bi-stable units or hysterons, which we denote as "+" or "−", are characterised by two reduced fields; h_C – a local coercive field, and h_U – an interaction field, these are illustrated in **Fig. 6(a)**. A switching field couple, which can be expressed by the plane (h_a, h_b), where $h_a = h_U + h_C$ and $h_b = h_U - h_C$, characterises a bistable unit and must conform to certain conditions. Representing the saturation magnetisation and saturation magnetic fields as M_S and H_{Sat}, respectively, when $H > H_{Sat}$, all bistable units are positive and the magnetisation will be $M = M_S$. At the other end of the hysteresis loop, if $H \leftarrow H_{Sat}$, all bistable units will be negative and $M = -M_S$. The above assumptions lead to the following general conditions for the couple, (h_a, h_b):

$$h_a \leq H_{Sat}$$

$$h_b \geq -H_{Sat}$$

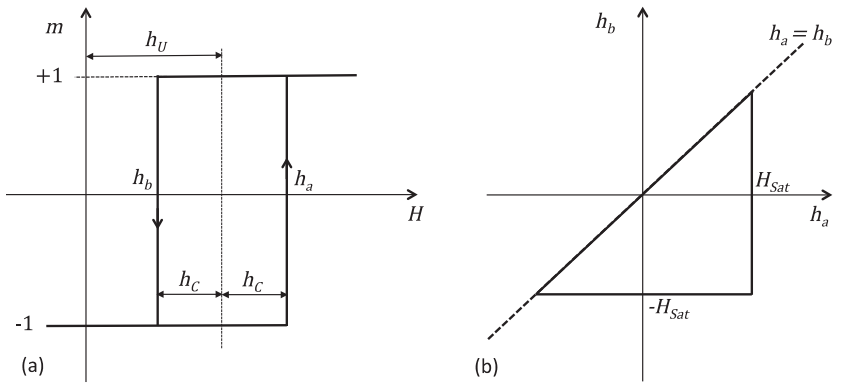

Fig. 6 (a) Elementary Preisach loop, with a local reduced coercive field, h_C, and an interaction field, h_U. (b) The Preisach plane (h_a,h_b) showing the triangle defined by the saturation fields.

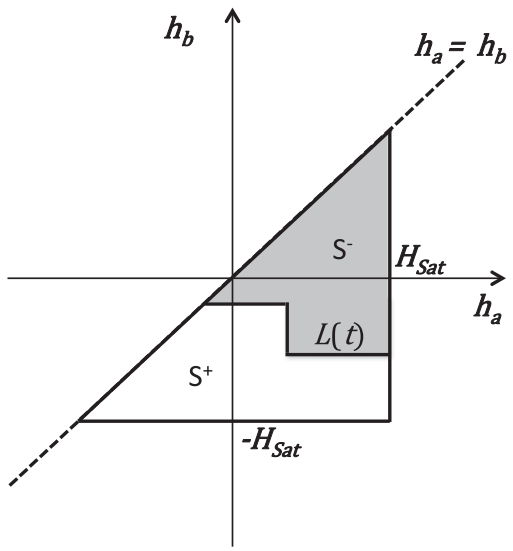

Fig. 7 The triangle defining the Preisach plane with the separation of the surfaces for the (−) and (+) states and the boundary between them as given by the line $L(t)$.

Given that hysteresis is an energetically dissipative phenomenon, we must have $h_a \geq h_b$. These three conditions allow us to define a triangle in the Preisach plane, as illustrated in **Fig. 6(b)**. Each couple for a particular bistable unit must belong to this plane.

The magnetisation process is determined by the Preisach density function, $p(h_a,h_b)$ of the material in question. Mayergoyz used a geometrical interpretation of the magnetisation process using a one-to-one correspondence between an operator $\hat{\gamma}_{a,b}$ and the points (h_a,h_b) in the Preisach plane for the half-plane $h_a \geq h_b$, (Mayergoyz, 1986a,b). The specific triangle, at any instant in time and depending on the sample history, can be split into two distinct regions: $S^{+(t)}$, which consists of points (h_a,h_b) for which $\hat{\gamma}_{a,b} = 1$, and $S^{-(t)}$, which consists of points (h_a,h_b) for which $\hat{\gamma}_{a,b} = -1$. This is schematically illustrated in **Fig. 7**. The boundary between the S^+ and S^- regions, defined as $L(t)$, is typically a stepped function whose vertices have h_a and h_b coordinates coinciding with local maxima and minima of input at the previous instant of time. The line $L(t)$ commences at the line $h_a = h_b$ and moves when the input changes. As such $L(t)$ represents the history of the sample. For example, if the magnetic field is decreased, the horizontal part of the line $L(t)$ moves downwards, while if the field increases, the vertical part of the line $L(t)$ moves to the right. Using this logic, the magnetisation state of the sample is evaluated via the following expression (see for example, Hejda and Zelinka, 1990; Mayergoyz, 2008a, b):

$$M = M_S \iint_{h_a \geq h_b} p(h_a, h_b) \hat{\gamma}_{a,b} dh_a dh_b \tag{10}$$

Eq. (10) can be rearranged in the following manner:

$$M = M_{Sat} \tag{11}$$

Therefore the magnetic state of the system is explicitly related to the broken line $L(t)$. As stated above, this is determined by the extremal values of input at previous instants of time. Consequently, the past extremal values of input shape the

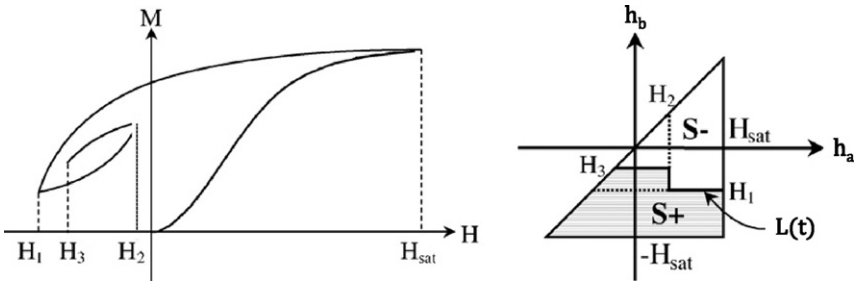

Fig. 8 Example of a magnetisation process in which the sample is taken from the virgin state to saturation in the positive sense, then the field is cycled from H_{Sat} to H_1, H_2 and H_3. Reprinted from Benabou, A., Clénet, S., Piriou, F., 2003. "Comparison of Preisach and Jiles-Athertonmodels to take into account hysteresis phenomenon for finite element analysis". J. Magn. Magn. Mater. 261, 139–160.

interface and thus leave a mark on the sample history, i.e., the nonlocal memory. These input values can be erased in what is referred to as the *wiping-out* property: Each local maximum wipes out the vertices whose h_a coordinates are below this maximum and each local minimum wipes out the vertices whose h_b coordinates are above this minimum. This can be incorporated into the memory vector h, which includes extrema, H_i, with coordinates that adhere to the following conditions (Benabou *et al.*, 2003):

$$H_0 = 0, h = \{H_0, H_1, H_2, H_3, \ldots, H_n\} \tag{12}$$

for $i=1,\ldots,n-1$ and $d_i=H_i-H_{i-1}$. Where we have:

$$\begin{aligned} d_i.d_{i+1} &< 0 \\ |d_{i+1}| &< |d_i| \end{aligned} \tag{13}$$

In Eq. (12) H_n, the last component, is the current value of the magnetic field. From these relations the memory vector is constructed. An example is illustrated in **Fig. 8** for the magnetic state defined by the memory vector $h=\{0,+H_{Sat},H_1,H_2,H_3\}$.

A further property of the model, known as the *congruency property*, states that all hysteresis loops (major or minor) corresponding to the same extremal values of input are congruent. This means that given a set of input variations within the same range, the output increments to the magnetisation must be the same.

The missing element is the knowledge of the Preisach density function. From this it is possible to determine the form of the hysteresis loop and more generally the variation $M(H)$ from any starting position within the hysteresis loop. This can be determined experimentally, as outlined by Mayergoyz (1986a,b), see also Hejda and Zelinka (1990). In general there is a good agreement between experiment and theory and modelling using the Preisach formalism provides a good insight into the magnetisation process. However, critics of the model state that the Preisach model, while being widely applicable on the macroscopic scale, does not always give a physical picture and only really allows a modelling of irreversible processes. Reversible processes must be introduced arbitrarily (Jiles, 2002). The relation between microstructure and magnetic properties is an essential aspect of understanding the magnetisation process. Often this can be investigated using computational micromagnetics, as illustrated by Dupré *et al.* (1999, 2002). A more generalized Preisach model includes an extension to the classical approach and describes the vector nature of the magnetisation process (Mayergoyz and Friedman, 1988; Hejda and Zelinka, 1990; Bertotti, 1991).

2.1.3 Jiles – Atherton model

The Jiles – Atherton or J – A model in its original presentation (Jiles and Atherton, 1986) aims at expressing the relation between the magnetisation of a ferromagnetic material as a function of the applied magnetic field. As a starting point, the model is based on the anhysteretic behaviour, $M_{an}(H)$, which can be expressed in the form of a modified Langevin function:

$$M_{an}(H) = M_{Sat}\left[coth\left(\frac{H_e}{a}\right) - \frac{a}{H_e}\right] \tag{14}$$

where $H_e=H+\alpha M$ is the effective field experienced by the magnetic moments within the ferromagnetic domains. H is the applied magnetic field and α is a mean field parameter analogous to the Weiss mean field. The parameter α has the dimensions of magnetic field and characterises the shape of the anhysteretic magnetisation. Typically this expression represents the effects of moment rotation within magnetic domains but does not account for losses due to domain wall motions. Jiles and Atherton (1986) note that for elevated values of the mean field constant, α, which translates as strong interactions between magnetic moment, hysteretic behaviour is observed, however, these values are unrealistic in practical terms.

Central to the J – A model is the consideration of the domain wall displacement through pinning sites, such as impurities and defects, where energy will be dissipated. The effects of both rigid, planar walls and flexible walls gives rise to irreversible and reversible magnetisation processes. Reversible processes can be thought of as the effect of wall bending between anchor points due to pinning sites. Irreversible processes are more associated with rigid wall motion. Energy losses due to hysteresis arise from the

irreversible changes, which are proportional to the average pinning energy:

$$\langle \epsilon_{pin} \rangle = \frac{\langle \epsilon_\pi \rangle}{2}(1 - cos\vartheta) \tag{15}$$

where $\langle \in_\pi \rangle$ denotes the pinning energy of a site for 180° domain walls and ϑ is the relative orientation of the magnetisation on either side of the domain wall. The pinning energy can be related to the change of magnetisation via (Jiles and Atherton, 1986): $E_{pin}(M) = k\int_0^M dM$, where $k = n\langle \in_\pi \rangle / 2m$, here n denotes the average density of the pinning sites in the solid and m the magnetic moment per unit volume. Assuming a uniform distribution of pinning sites, where each is considered as having the mean pinning energy, the work done against pinning by the applied field will be proportional to the change in magnetisation. The magnetisation energy $\int M dB_e$, with $B_e = \mu_0 H_e$, will be the difference in energy between that obtained in the ideal lossless case, i.e., $\int M_{an}(H_e) dB_e$, minus the loss due to hysteresis, $k\int dM$:

$$\int M dB_e = \int M_{an}(H_e) dB_e - k\int \left(\frac{dM}{dB_e}\right) dB_e \tag{16}$$

From which it is a simple matter to obtain:

$$M = M_{an}(H_e) - \delta k\left(\frac{dM}{dB_e}\right) \tag{17}$$

The parameter δ has a value of $+1$ for $dH/dt > 0$ and -1 for $dH/dt < 0$, ensuring that the pinning opposes changes to the magnetisation. A consideration of the magnetic energy under the assumption of a uniform distribution of pinning sites leads to conclusion that the magnetisation energy will be the difference between the energy which would be obtained in the anhysteretic case minus the energy due to the losses induced by domain wall motion. It is then possible to obtain the differential susceptibility for the irreversible magnetisation, M_{irr}, in the form:

$$\frac{dM_{irr}}{dH} = \frac{M_{an} - M_{irr}}{k/\delta\mu_0 - \alpha(M_{an} - M_{irr})} \tag{18}$$

The reversible contribution to the magnetisation process is linked with the anhysteretic and irreversible magnetisation via the linear relation:

$$M_{rev} = c(M_{an} - M_{irr}) \tag{19}$$

where the reversibility coefficient, c, determines the proportion of the magnetisation that is reversible and can take values between zero and unity. Given that the total magnetisation can be expressed as the sum of reversible and irreversible components; $M = M_{rev} + M_{irr}$, and taking into account Eq. (19), we can express the magnetisation as $(1+c)M = M_{irr} + cM_{an}$ and the total differential susceptibility takes the form:

$$\frac{dM}{dH} = \frac{1}{(1+c)}\frac{M_{an} - M_{irr}}{k/\delta\mu_0 - \alpha(M_{an} - M_{irr})} + \frac{c}{(1+c)}\frac{dM_{an}}{dH} \tag{20}$$

This equation can then be used to derive the magnetisation as a function of the magnetic field. An alternative form of this equation can be expressed as (Tumanski, 2011):

$$\frac{dM}{dH} = \frac{M_{an} - M_{irr}}{k/\delta\mu_0 - \alpha(M_{an} - M_{irr})} + c\left(\frac{dM_{an}}{dH} - \frac{dM_{irr}}{dH}\right) \tag{21}$$

The J–A model has a number of parameters that require adjusting to obtain the correct shape for the hysteresis loop, these include k,c,a and α, which in general can be determined from the experimental measurement.

Optimisation of the fitting parameters can be computed and once correctly determined provide the model with excellent agreement to experimentally determined $M(H)$ measurements. An example is illustrated in **Fig. 9**. The importance of the model is that the fitting parameters have a specific physical meaning and that it provides a useful tool for discussing the physical mechanisms of the magnetisation process.

It is worth noting that under certain conditions (Pasquale *et al.*, 1998), the Preisach and the J–A models are equivalent. In particular, if the Preisach distribution function decreases exponentially with h_c, according to the coercive force of the Preisach formalism, then the rate of change of the magnetisation with the applied magnetic field in the two models will be equivalent. Modified forms of the Jiles–Atherton model have been developed; for example, Hamimid *et al.* (2010) introduce the magnetisation, M, in place of the irreversible magnetisation, M_{irr}, into the effective magnetic field, H_e. The authors demonstrate that this allows a better fitting of their experimental magnetic hysteresis loops.

2.1.4 Globus–Guyot model

The original model ascribed to Globus considers the initial magnetisation of soft magnetic materials, which is due to the motion of a domain wall, as opposed to the domain rotation in hard magnetic materials at high magnetic fields. Domain wall motion can occur via the wall bowing mechanism or wall displacement. The bowing of domain walls is considered in an analogous manner to the elastic deformation of a membrane, which is fixed, or pinned, by a grain boundary. The domain wall is subject to the force created by an applied magnetic field which produces domain growth of those domains with a favourable direction of magnetisation with respect to the orientation of the applied magnetic field. Such motion, since it is only due to bowing, is reversible. This

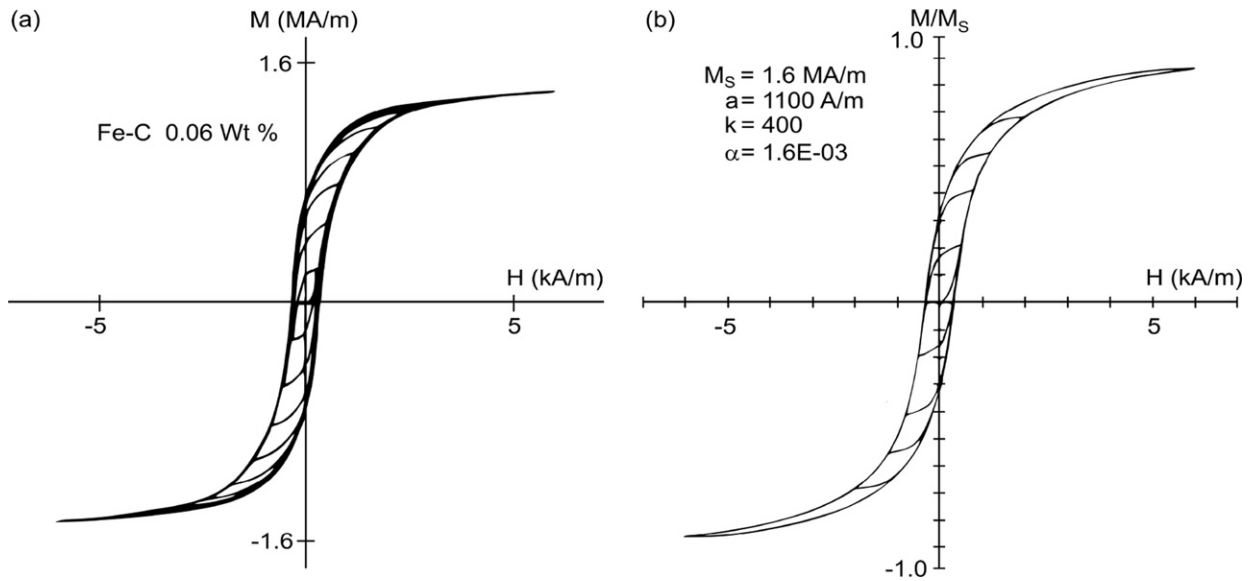

Fig. 9 (a) Experimental magnetic hysteresis loops for a sample of Fe-C 0.06 wt. %. (b) Theoretical magnetic hysteresis loops for the J–A model with $M_{Sat}=1.6\times10^6$A/m; $a=1100$ A/m; $a=1.6\times10^{-3}$ and $k=400$ as determined from Eq. (18). The reversible portions were determined using Eq. (20). Reprinted from Jiles, D.C., Atherton, D.L., 1986. "Theory of ferromagnetic hysteresis". J. Magn. Magn. Mater. 61, 48–60.

means that if the magnetic field is reduced the bowing will correspondingly reduce, with the domain wall following the same path that it took with increasing field. Should the magnetic field be sufficient to cause the depinning of the domain wall from the pinning site, then domain displacement will occur and the process is irreversible, with energy being dissipated and lost to the domain wall. Initial work on pinning mechanisms was proposed by Becker and Döring (1939) and Kersten (1943).

As stated above, the initial model (Globus, 1962) considered the reversible portion of the motion of domain walls in ferrimagnetic materials, discussing the relation between the initial susceptibility and the domain wall continuity, which is dependent on sample crystallinity (grain size) as well as the relative orientations of the crystalline axis with respect to the domain wall direction. Globus and Duplex (1969) also considered the effect of temperature, with wall continuity reducing with temperature. Experimental results were well explained using the model, which shows that the diameter of the wall inside the grains controls the susceptibility of the material (Globus and Duplex, 1970). The authors also indicate that wall motion dominates domain rotation, which are dependent on anisotropy but not grain size, D_m. Furthermore, the model predicts a magnetisation that is proportional to the applied field and the grain size. Beyond a critical value of the applied magnetic field, $H_C\propto 1/D_m$, the wall becomes detached from the pinning site and moves. This leads to a reduction in the size of the domain wall area (Globus and Guyot, 1972). For fields above this critical value, the position of the domain wall is determined by a balance between forces on the wall produced by the applied field, which are proportional to d^2, and the viscous forces proportional to d, where d is the wall diameter (for a spherical grain) and is proportional to $1/H$ (Globus *et al.*, 1971). Globally the model presents a schematic hysteresis loop as illustrated in **Fig. 10**, where a domain wall is considered under the differing applied fields and directions and responds accordingly. Further considerations for the demagnetising field produce a tilting of the loop.

The concept of domain wall bulging and displacement are in general accord with the Jiles–Atherton model, though the latter consider that this is too restricted and that domain wall pinning also occurs on other inhomogeneities within the grains, such as tangles of dislocations, regions of inhomogeneous strain and other precipitates or non-magnetic inclusions (Jiles and Atherton, 1986).

The model was extended to include dissipative processes resulting from the translation of domain walls at more elevated values of applied field (Globus and Guyot, 1972; Guyot and Globus, 1973). For a polycrystalline material, the global anisotropy is considered to be composed of the magneto-crystalline contribution, K_1, as well as a magnetoelastic term, such that $K=K_1+\lambda_S\sigma_n$, where σ_n denotes the natural stresses within the material and depend on the sample composition. This extension of the model also considers the energy loss mechanism during wall displacement, which contributes to the irreversible mechanisms of the $M(H)$ loop. A simple model is presented of the displacement of the domain wall in a spherical grain, of radius R and saturation magnetisation M_S. In the remnant state, where the domain wall is flat, i.e., a circular membrane, the magnetisation will take a value given by:

$$M_R = 2M_S\frac{dV}{V} \tag{22}$$

where dV is the volume due to the wall displacement and V is the volume of the grain. With respect to **Fig. 11**, we can envisage the remnant magnetisation as the difference of the magnetisations of the two domains separated by a domain wall, of negligible thickness. The problem is reduced to a question of geometry and allows the remnant magnetisation to be expressed as $M_R=M_S(V_\uparrow-V_\downarrow)$. However, $V_\uparrow=V-V_\downarrow$, from which, using the volume formulae; $V=4\pi R^3/3$ and $V_\downarrow=\pi h^2(3R-h)/3$, with

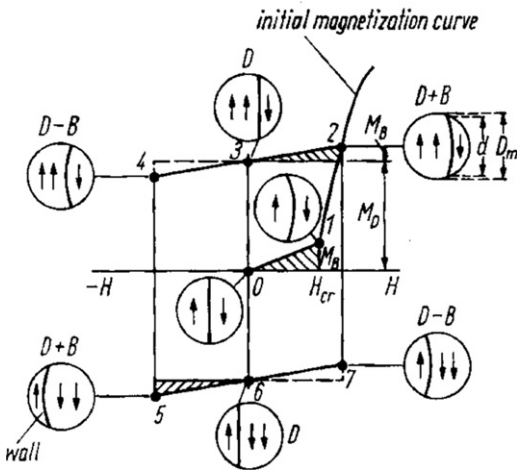

Fig. 10 The Globus–Guyot model for a spherical grain showing the roles of domain wall bulging and displacement. Hatched areas denote the magnetisation due to the bulging of the domain wall. Reproduced from Globus, A., Guyot, M., 1972. "Wall displacement and bulging in magnetisationmechanisms of the hysteresis loop". Phys. Stat. Solidi (b) 52, 427–431.

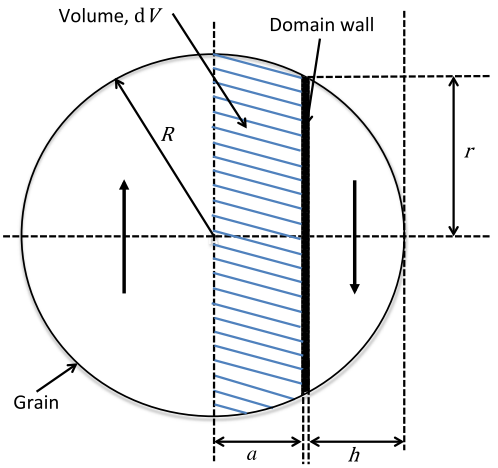

Fig. 11 Remnant state magnetisation, M_R, according to the Globus model, where a single flat domain wall in a spherical grain is displaced from the centre by a distance a.

$h = R - a$, it is possible to express the remnant magnetisation in the form:

$$M_R = \frac{M_S}{2}\left[3\frac{a}{R} - \left(\frac{a}{R}\right)^3\right] \tag{23}$$

The pinning force on the domain wall is expressed as $F = 2\pi r f$, where f is the pinning force per unit length. F is considered to be a frictional force acting to resist the motion of the domain wall under the action of an applied field. The work done on the wall during a cycle (considered as the movement of the wall between positions a_0 and $-a_0$) is given as (Guyot and Globus, 1973, 1977):

$$W_{pin} = 4\int_0^{a_0} F da = 8\pi f \int_0^{a_0} r da \tag{24}$$

As the wall displaces there will be a variation of its total energy due to the variation of the wall area, ΔS. This leads to an energy loss during the cycle, which is expressed in the form:

$$W_{loss} = 4n\gamma\Delta S = 4n\pi\gamma\left(R^2 - r_o^2\right) = 4n\pi\gamma a_0^2 \tag{25}$$

where γ denotes the wall energy per unit area and n is a constant of proportionality. The total loss energy can then be expressed as:

$$W = W_{pin} + W_{loss} = 4\pi\left(2f\int_0^{a_0} r da + n\gamma a_0^2\right) \tag{26}$$

Under the conditions of weak magnetisation the energy per unit volume can then be expressed in the simplified form:

$$E = \frac{3a_0}{R^2}\left(2f + n\gamma\frac{a_0}{R}\right) = \frac{4}{R}\frac{M_R}{M_S}\left(f + \frac{n\gamma}{3}\frac{M_R}{M_S}\right) \tag{27}$$

where we have used the substitution: $a_0/R = 2M_R/3M_S$. The model expressed by Eq. (27) is in good agreement with experimental results. Experimental results can then be used to determine the domain wall energy, $\gamma = 2\sqrt{AK_1}$ and f, (Guyot and Globus, 1973, 1977). If the anisotropy constant, K_1, is known, then knowledge of the domain wall energy permits the evaluation of the exchange constant, A.

2.2 Magnetic Domains, Domain Walls and Hysteresis

In the above section we have outlined some of the principal ingredients for the theoretical understanding of magnetic hysteresis. In this section we will briefly discuss the relation between the hysteresis phenomenon and the existence of magnetic domains, including domain walls. This will complete some of the elements that were presented in the previous section. This will naturally take into account some of the energy considerations that we have outlined above.

A magnetic domain wall is a region of transition in which the atomic magnetic moments vary smoothly in orientation from one magnetic domain to a neighbouring domain. When the magnetic moments of the neighbouring domains are anti-parallel, the domain wall is referred to as a 180° wall. There are two principal types of domain wall; the *Bloch wall* is characterised by the variation of the orientation of the magnetic moment in a direction perpendicular to the direction of transition or in the plane of the wall; the *Néel wall*, on the other hand, concerns a rotation of the moment orientation in the direction of transition or perpendicular to the domain wall. In either case the domain wall represents an additional energy contribution due to the excess energy associated with neighbouring magnetic moments which are non-collinear and hence have higher exchange energies and also a contribution due to the fact that the moments are not aligned with the easy axis as defined by the magnetocrystalline energy. The energy per unit surface area in say the y–z plane, where the transition direction is parallel to the x-axis, can be expressed as:

$$\gamma_W = \int_{-\infty}^{\infty} [u_{exch}(x) + u_{An}(x)]dx \tag{28}$$

where the first and seconds are respectively the exchange and magnetic anisotropy energies. The wall energy expressed in Eq. (28) requires the explicit form of the contributions for the exchange and anisotropy terms, which will depend on the type of sample and the type of domain wall. By way of example we consider the case for a 180° Bloch wall for a sample with uniaxial anisotropy in zero applied magnetic field. In this case we can express the exchange energy in the form (Cullity, 1972):

$$u_{exch} = -2JS_i \cdot S_j = -2JS_iS_j cos\vartheta \tag{29}$$

where J is the exchange integral, S_i and S_j are the spin angular momenta for neighbouring atoms i and j, respectively and ϑ denotes the angle between these neighbouring spins. Allowing for the case where the spin momenta are equivalent, $S_i = S_j = S$, and using the series expansion for $cos\vartheta = 1 - \vartheta^2/2 + \vartheta^4/24 - \ldots$, we can express the exchange energy as:

$$u_{exch} = JS^2\vartheta^2 - 2JS^2 \tag{30}$$

where we have limited the range of the series taking into account that ϑ is very small. Since the second term has no dependence on angle, we can neglect it, as it will be that same for regions within a domain as within a wall. It is the first term which expresses the excess energy due to the existence of the domain wall itself. For a wall thickness of N atoms, which are separated by a distance a, we can express the excess exchange energy per unit area of the wall:

$$\gamma_{exch} = JS^2\vartheta^2\frac{N}{a^2} \tag{31}$$

Since we are considering a 180° domain wall we can write: $\vartheta = \pi/N$, which yields:

$$\gamma_{exch} = \frac{JS^2\pi^2}{Na^2} \tag{32}$$

We now consider the anisotropy contribution to the energy from Eq. (28). In the simplest case we can expect that the excess energy will be related to the anisotropy constant K times the wall volume, leading to the anisotropy term:

$$\gamma_{an} = KNa \tag{33}$$

Summing these contributions and writing the wall width as $\delta = Na$, we obtain:

$$\gamma = \frac{JS^2\pi^2}{\delta a} + K\delta \tag{34}$$

Minimising the wall energy with respect to the wall width allows us to obtain the relation for the wall width in terms of the exchange stiffness constant; $A = JS^2\pi^2/a$ and the anisotropy constant:

$$\delta = \sqrt{\frac{A}{K}} \tag{35}$$

A more refined calculation of the domain wall width gives a value which is somewhat larger: $\delta = \pi\sqrt{A/K}$, Skomski (2008). Substituting the result from Eq. (35) into (34) then yields a total domain wall energy of:

$$\gamma = 2\sqrt{AK} \tag{36}$$

The more refined approach yields a value which is twice this value, Skomski (2008). The competing energy terms illustrate that the magnetocrystalline anisotropy favours narrow domain walls, since this would align more spins along the easy axis directions and reduce the excess anisotropy energy. However, narrow walls mean that within the domain walls the magnetisation gradients would be large and unfavourable for the exchange energy, which is reduced when the angle between neighbouring spins is small. It is therefore the ratio of these quantities that determines the extent of the transition region between magnetic domains. For example, the domain wall thickness for Fe is 64 nm with a wall energy of 4.1 $\mathrm{mJm^{-2}}$ (Coey, 2009).

The phenomenon of hysteresis is intimately linked with the microscopic processes occurring in a magnetic body and concerns the manner in which the magnetisation is reversed due to the application of an external magnetic field. This relates not only to the direction in which the field is applied but also to the specific magnetic state of the sample, which depends on the magnetic history and the particular configuration /distribution of the magnetic spins. Also of importance will be the purity and crystallinity of the sample itself. Impure samples and grain boundaries will have a significant influence on the manner in which domain walls are nucleated and how they propagate with the motion of domain walls. We have already touched on some of these issues in the preceding sections.

There are several modes of magnetisation reversal, which include coherent rotation, curling and multi-domain reversal. This latter is associated with hysteresis losses and is most common for magnetic bodies with dimensions greater than the single domain particles. The critical size can easily be calculated from energy considerations and for a spherical body of radius R_{sd} can be expressed as, (Hubert and Schäfer, 1998):

$$R_{sd} \approx 36\frac{\sqrt{AK}}{\mu_0 M_s^2} \tag{37}$$

with values ranging from a few nm in soft magnets to around a micron in hard materials, Skomski (2008). The critical moment of any reversal process is the nucleation, which for the multi-domain mechanism corresponds to the creation of a reverse domain typically due to a defect at the surface of the sample. Once nucleation has taken place, the motion of the domain wall under the action of a reverse applied field requires very little energy and leads to the expansion of the reversal domain unless pinning of the domain wall occurs. Domain sizes and shapes can be rather difficult to predict, and will depend on a number of factors, such as the exchange energy, anisotropy, sample dimensions, sample shape and defects. Micromagnetic simulations have been quite successful in many cases for small magnetic structures, however, larger structures are more difficult to predict (Hubert and Schäfer, 1998).

In general domain walls tend to be flat or planar, due to energy minimisation of magnetostatic and exchange energies. However, non-planar walls can exist due to pinning of domain walls at inclusions, voids and internal stresses in the sample. Strong pinning tends to occur on planar defects, especially those with dimensions comparable to the domain wall width, δ. Large planar defects can be very effective pinning centres since they will tend to reduce the wall area and hence reduce the total energy or excess energy of the system. Weaker pinning can be expected for smaller defects. Typically a sample will contain a number of defects of varying sizes and will be distributed throughout the body of the magnetic entity. Indeed, pinning is the principal source of hysteresis and coercivity in inhomogeneous magnets. We can think of pinning as the domain wall being trapped in a region with low domain wall energy, and the coercivity is determined by the "depinning" field, which is the magnetic field necessary to push the wall over the pinning energy barrier.

A simple model of wall pinning considers a planar wall, with position x, in which the energy of the wall depends on position. The coercivity is evaluated from the total wall energy, which can be expressed in the form, (Skomski, 2008):

$$E_W(x) = \eta(x) - 2\mu_0 M_S H x \tag{38}$$

where $\eta(x)$ depends on the microstructure and is essentially proportional to the local wall potential and the second term is a Zeeman energy. For a given applied field, H, the wall position can be deduced from the minimisation of the energy, $E_W(x)$. The field can be expressed from this as: $H = \frac{1}{2\mu_0 M_S}\frac{d\eta}{dx}$. The derivative $d\eta/dx$ is a function of position and will have a maximum in the vicinity of a defect. Increasing the field at this point will push the domain wall over the barrier and lead to its depinning. The depinning field can thus be expressed as:

$$H_P = \frac{1}{2\mu_0 M_S}\left(\frac{d\eta}{dx}\right)_{max} \tag{39}$$

The passage of the domain wall through a defect is typically envisaged as the progress of the wall bulging from the defect position until the field is sufficient to depin it, resulting in a jump of the wall to a new position and returning to its planar state. This mechanism of domain wall pinning was first presented by Kersten (1943), see also work by Gaunt (1983, 1986). These wall jumps are associated with the discontinuities in the magnetisation, known as *Barkhausen jumps*. These jumps represent the irreversible movements of the domain wall and hence the magnetisation, leading to coercivity and the phenomenon of hysteresis. Indeed the jumps can be heard as Barkhausen noise with clicks and illustrate the dissipation of energy in the system. This has lead to the study of magneto-acoustic emission in ferromagnetic and ferrimagnetic materials. In addition to the pinning mechanism, Guyot and Cagan (1991) also suggest that magnetic hysteresis is more fully described

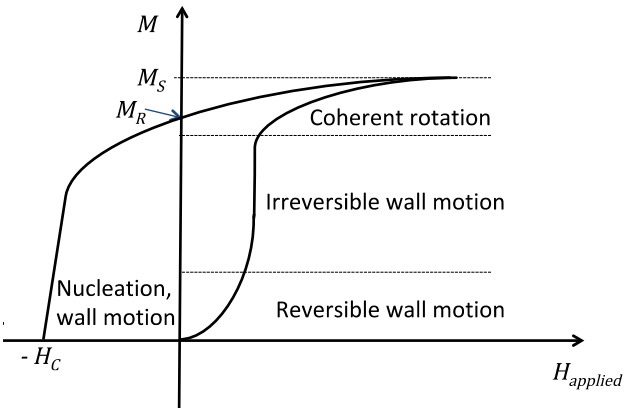

Fig. 12 Schematic illustration of the initial magnetisation curve and demagnetisation in the first and second quadrants.

by the addition of domain wall creation and annihilation, which was shown to be a dissipative process (Guyot and Globus, 1973, 1977). Guyot and Cagan (1991) suggest that the physical mechanism of magnetic energy conversion into heat arises from the acoustic emission process.

We can summarise the magnetisation process from the virgin state to saturation and back to zero field and then to the first reverse coercive field. This allows an overall view of the principal mechanisms at play. This magnetic path can be schematically represented as illustrated in **Fig. 12**. In the first stages, magnetisation increases as domain walls move such that those domains that are favourable with the direction of the applied field grow. Here the wall can be pinned by defects and can bulge or bow. However, since this process is reversible, any reduction of the applied field results in reduction of the magnetisation along the same path back to the virgin state. If the field is further increased, Barkhausen jumps can occur as domain walls over come the energy barrier created by pinning sites and the magnetisation increases with the increase of the favourable domain orientation. This region is irreversible since the erratic jumps dissipate some of the energy and a reduction of the field from this point will result in a reduction of the magnetisation along a different path and will eventually lead to some remnant magnetisation at zero field. As the magnetic field is further increased, the non-favourable domains are eliminated from the sample and we enter the reversible coherent rotation phase, also referred to as the approach to saturation, where the magnetisation rotates towards full alignment with the applied magnetic field. This leads to full saturation of the sample. When the field is reduced, reverse domains can nucleate and begin to propagate as the sample returns to a multidomain state. At zero field, the remnant magnetisation is attained, where there is still an overall magnetisation in the direction of the previously applied field. Applying a magnetic field in the reverse direction will increase the domains in the reverse direction until a state of zero magnetisation is reached at the coercive field. This microscopic state of the sample is very different from the virgin state, which can now only be re-established by heating the sample to a temperature above the Curie point and then cooling in zero field.

3 Experimental Methods to Measure Magnetic Hysteresis

The objective of the present section is to discuss the underlying concepts behind the most common and most popular experimental methods used to characterise magnetic materials by means of their hysteresis curves. Magnetometry techniques for determining hysteresis loops and magnetisation as function of an applied magnetic field can be classified as closed-circuit or open-circuit measurements. In open-circuit measurements, the sample is not part of a complete magnetic circuit and due to demagnetisation effects, the externally applied field H' is different from the internal field H of the sample. The relationship between both quantities is given by $H = H' - NM$, where N is a demagnetising factor which depends on the shape of the sample. Open-circuit methods can further be divided into two categories, force- and induction-based, where either the force acting on a magnetised sample in a magnetic field gradient is measured or the change of magnetic flux in a circuit is sensed while the sample is moved, respectively. More specific techniques are based on optical effects such as the magneto-optical Kerr effect (MOKE). In closed-circuit techniques, the sample is part of a complete magnetic circuit and in the obtained $B(H)$ or $M(H)$ loops, the applied field H' and the internal field of the sample H are identical. A common example is the hysteresisgraph used to characterise permanent magnets. The principal open-circuit techniques used nowadays include the vibrating sample magnetometer (VSM) and the alternating gradient force magnetometer (AGFM). When ultimate sensitivity is paramount, a superconducting quantum interference device (SQUID) magnetometer is the method of choice. For fast measurements of the $B(H)$ dependence without the need to know the absolute value of M, either a B-H looper or a MOKE magnetometer can be used. In the following, we will briefly discuss these five open-circuit techniques.

3.1 B-H Looper

An ac hysteresis loop tracer, or B–H looper, typically consists of a pair of Helmholtz coils, capable of generating ac magnetic fields of up to 150 mT at frequencies in the Hz to lower kHz range. The sample is placed in the centre of a pair of pick-up coils, which are

connected such that the voltages induced by the changing flux due to the applied ac field are cancelled out, whereas the voltages induced by the changing magnetic flux, Φ, of the sample are enhanced. According to Faraday's law of induction, the voltage V_i induced in the pick-up coils is given by:

$$V_i = N\frac{\partial \Phi}{\partial t} = NA\frac{\partial B}{\partial t} \tag{40}$$

where N is the number of turns of the pick-up coils and A is the cross-sectional area. The time integral of the above equation yields the value of the magnetic flux density B. The value of the magnetic field H can be measured either by a Hall probe attached to one of the pick-up coils inside the Helmholtz coils or from the voltage across a resistor connected in series to the Helmholtz coil.

B-H loopers are cheap and simple and are typically used for quickly characterising soft magnetic films by measuring their $B(H)$ loops, but due to the small fields generated by the Helmholtz coils, their application is limited to thin films with small coercivity. Moreover, relatively large sample volumes are required due to the low sensitivity, and absolute magnetic moment measurements require difficult calibration.

3.2 Superconducting Quantum Interference Device Magnetometer

SQUIDs are the most sensitive devices available today for detecting the magnetic flux Φ or any quantity that can be transformed into it. Their operation is based on two physical phenomena: flux quantisation in a closed superconducting loop in units of the flux quantum $\Phi_0 = h/2e = 2.07 \times 10^{-15}$ Tm^2, and Josephson tunneling. The key part of any SQUID sensor, which is depicted in **Fig. 13(a)** and **(b)**, is a Josephson junction which separates two superconductors by means of a "weak link." The most important type of barrier used nowadays is a Josephson tunnel junction made from an insulator such as AlO_X. If the junction is biased with a constant current I_b, the operational point of its I–V curve lies midway between superconducting and normal behaviour. Since the current–voltage characteristic of a Josephson junction exhibits a hysteresis itself, an additional shunt resistor is placed across the junction to dampen its dynamics (Fagaly, 2006). When magnetic flux is inductively coupled into the SQUID loop, screening currents are generated which effectively increase or decrease the critical current I_c of the junction, that is given by the first Josephson equation as $I_c = I_{c0}\, sin\varphi$ with $\varphi = \Phi_1 - \Phi_2$ being the phase difference across the junction (Josephson, 1962). I_{c0} is the maximum critical current of the junction and depends on the thickness of the insulating barrier, the junction area as well as the operating temperature. When biased with $I_b > I_{c0}$, the phase difference across the junction changes over time and the voltage drop across the junction will vary. This is described by the second Josephson equation (Josephson, 1962):

$$\frac{\partial \varphi}{\partial t} = \frac{2e}{\hbar} \cdot V_{DC} = \frac{2\pi}{\Phi_0} \cdot V_{DC} \tag{41}$$

where $\hbar = 1.055 \times 10^{-34}$ Js is the reduced Planck's constant. As shown in **Fig. 13(b)**, an increase or decrease of external magnetic flux leads to periodic changes of the voltage V_{DC} with the corresponding period being that of the flux quantum Φ_0.

Different types of SQUIDs are available that differ in the manner of biasing the Josephson junction as well as in the number of junctions itself (Fagaly, 2006). In dc SQUIDs, two junctions are biased by a dc current, whereas in rf SQUIDs, there is only one Josephson junction which is biased by a rf current. Due to their generally superior noise performance, nearly all commercial SQUIDs are dc SQUIDs. For use as a magnetometer, a magnetic thin film or wire wound pick-up coil can be connected to the input coil of a current sensor SQUID. Other applications include the use as magnetic gradiometers, current sensors and voltmeters, as well as amplifiers (Schmelz and Stolz, 2017). While an unparalleled sensitivity of 10^{-10} Am^2 or better can be attained, the measurements are slow and time-consuming since they involve both data extraction and changing the field of the superconducting magnet from one measuring point to the next during a hysteresis cycle.

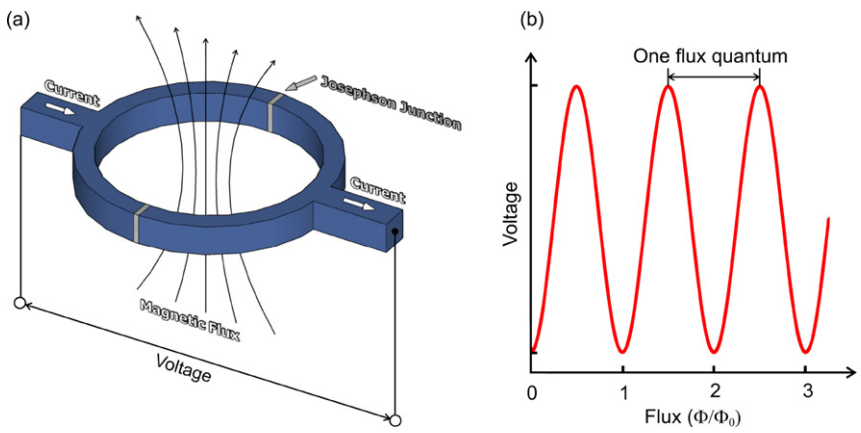

Fig. 13 (a) dc SQUID with two weak links (Josephson junctions) and (b) oscillations of the voltage across a Josephson junction as a function of the magnetic flux.

3.3 Vibrating Sample Magnetometer

The VSM, developed by Foner (1959) and schematically drawn in **Fig. 14**, is one of the most commonly used instruments in academic and industrial research to measure the magnetic moment not only of soft and hard magnetic materials, but also of weakly magnetic or paramagnetic samples. Its operating principle is based on attaching the sample to a vibrating rod located in the centre of a single or double set of static pick-up coils. The exact coil arrangement and direction of sample vibration relative to the external field depend on the type of magnetic field source used. If a vertical field is applied, as with a superconducting solenoid, a pair of Helmholtz coils together with longitudinal sample vibration is employed. When a horizontal field is applied, either with an electromagnet or by permanent magnet sources such as, e.g., Halbach cylinders (Halbach, 1980), an assembly consisting of, e.g., a four-coil configuration known as Mallinson's set (Mallinson, 1966) in conjunction with perpendicular sample vibration is used. In either case, upper and lower coils are oppositely wound such that the signals induced in them by the oscillating sample are of additive nature, whereas any signal resulting from fluctuations of the external field is minimised or ideally cancelled. The voltage V_i generated in the pick-up coils is proportional to the magnetisation of the sample and depends on the orientation of the magnetic moment relative to the coils. For a magnetic moment m directed along the (x-)axis of a coil pair and moving with a velocity $\dot{x}$ the induced voltage V_i is given by (Fiorillo, 2010):

$$V_i(x,t) = \frac{d}{dx}\left(mk_{(x)}(x)\right)\cdot\dot{x} = mg_x(x)\cdot\dot{x} \tag{42}$$

Here, $g_x(x) = dk_x(x)/dx$ represents the distinct spatial sensitivity distribution of the coil arrangement, and thus the dependence of the VSM signal on the sample position (Zieba and Foner, 1982). To obtain time dependence of the signal, the above expression must be integrated over the volume of the sample for given amplitude and frequency of oscillations. Ideally, sensing coil arrangements are used that exhibit a saddle point in the symmetry centre in whose vicinity the sensitivity is independent of the

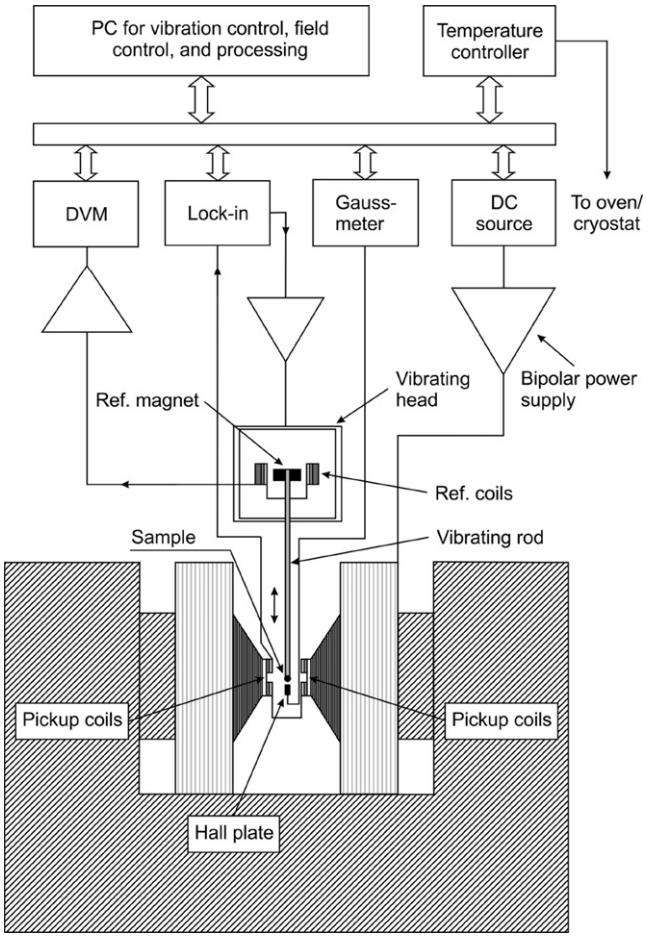

Fig. 14 Schematic drawing of a VSM with an electromagnet field source. The voltage induced in the pick-up coils is amplified using a lock-in amplifier. Its reference signal is used to control both amplitude and frequency stability of the oscillations of the vibrating rod. Reproduced from Fiorillo, F., 2010. "Measurements of magnetic materials". Metrologia 47, S114–142.

sample position. For example, for thin coils spaced apart by a distance of $d = \sqrt{3}\,a$, with a being the coil radius, the sensitivity function is not maximal, but very flat at the centre of the coil arrangement, and the linearity of the field derivative is enhanced.

The measurement of the magnetic moment of the sample can be performed by either varying the applied magnetic field continuously with time or by generating the magnetic field in steps. The latter offers an improved signal-to-noise ratio if the signal is averaged over a reasonably long integration time of the order of 1 s or longer. Typical lower and upper detection limits for the magnetic moment in commercially available VSMs are of the order of 10^{-9} Am^2 and 1 Am^2, respectively. The vibration frequency generally ranges from 10 Hz up to several 100 Hz, with the vibration amplitude varying between 0.1 and several tens of millimeters (Graham, 2000). Frequency stability and oscillation amplitude are controlled by a lock-in amplifier-based feedback loop. The volume of the sample is generally chosen to be a few cubic millimeters such that it does not protude from the region of uniform sensitivity determined by the sensing coil configuration. While any kind of sample shape, even irregular ones, can be tested, its influence on the measured hysteresis curve has to be properly taken into account by calculating the corresponding demagnetisation factors. The ideal sample geometry is that of a sphere, as it not only ensures uniform magnetisation of the sample, but also allows for precise and easy retrieval of the actual hysteresis curve after correction for the demagnetising field $N_d = 1/3$. Magnetic thin films can conveniently be approximated by the disc geometry with the corresponding demagnetisation factors being similar to those of an oblate ellipsoid having the same axes length. In any case, the sample has to be centred prior to a measurement in order to determine the saddle point of the sensitivity function.

When samples with very small volumes such as thin films are measured, it is typically necessary to subtract the diamagnetic contribution from the sample holder or substrate, which manifests itself in a negative slope of the entire hysteresis curve. Properly calibrated, VSM measures the absolute value of the magnetic moment of the sample from which, when divided by the sample volume, the saturation magnetisation can be determined. However, since an absolute calibration by numerically calculating the sensitivity function g is difficult, it is much more common to calibrate instead by comparing to the well-known magnetic moment of a standard sample such as a Ni sphere measured at a high field value. As an additional benefit, this type of calibration offers a correction from the image effect, which shows up as an apparent decrease in magnetisation at high fields due to saturation of the soft magnetic pole shoes of the electromagnet.

3.4 Alternating Gradient Force Magnetometer

An AGFM can be thought of as sort of an inverted VSM. Its original idea was developed by Zijlstra (1970), whereas first complete instruments were described by Reeves (1972), Roos *et al.* (1980), and Flanders (1988). In **Fig. 15**, the side view of an AGFM is shown. A magnetic sample attached to one end of a cantilevered quartz or glass fibre is magnetised by a dc field of varying amplitude and simultaneously subjected to a small ac field gradient. The alternating field gradient, whose maximum is of the order of several hundreds of mTm^{-1} at the location of the sample, exerts an alternating force on the sample, that is proportional to both the amplitude of the field gradient and the magnetic moment of the sample. For example, in a vertical gradient AGFM, a sample with its magnetic moment directed along the x-axis would be subjected to an alternating force $F_Z = \mu_0 m(\partial H_x/\partial_Z)$ in z-direction. The resulting deflection of the rod is converted into a voltage by means of a piezoelectric bimorph mounted to the other end of the rod. By operating at or close to the mechanical resonance frequency of the cantilever, typically between 100 and 1000 Hz, the corresponding output signal is greatly enhanced (Graham, 2000).

Compared to a VSM, the large Q factor of the mechanically resonant system directly translates into a gain in sensitivity by a factor of 50–100 (Graham, 2000). However, maximum measurable sample size, mass and moment are reduced which largely compensates this advantage. Factoring in mechanical and acoustic noise from the environment, a sensitivity of 10^{-10} Am^2 is more realistic. When measuring soft magnetic materials, the changing field gradient can lead to errors in the measurements of remanence, coercivity, and magnetisation decay.

3.5 Magneto-Optical Kerr Magnetometer

Optical magnetometers measure the magnetisation by means of magneto-optical effects such as the Faraday (1846) and Kerr (1877) effect and are especially useful in the case of thin films whose signals might be too weak for inductive or force-based methods. What is exploited for optical magnetometry are their respective rotational effects, both of which can be interpreted as circular magneto-optical birefringence causing a magnetisation-dependent polarisation rotation of light upon reflection from or transmission through a magnetic sample. The Faraday effect occurs in optically transparent samples, whereas the Kerr effect applies to any metallic or otherwise light-absorbing magnetic material with a sufficiently smooth surface. The latter is also much weaker than the transmission effect since the light only interacts with the magnetisation of a relatively thin surface layer whose thickness is given by the penetration depth of the light. Both effects are linear in magnetisation and consequently, the corresponding rotation is inverted when the magnetisation direction is reversed.

The physical origin of both effects is the interaction of light as an electromagnetic wave with the magnetisation-dependent properties of a sample. Linearly polarised light impinging upon a reflective magnetic surface will induce electrons to oscillate parallel to its plane of polarisation, which is the plane of the electric field, E. A fraction of the incident light beam is regularly reflected, N, i.e., it maintains its original polarisation. At the same time, the Lorentz force induces of small component of vibrational motion perpendicular to both the magnetisation and the primary motion. This secondary motion, which is

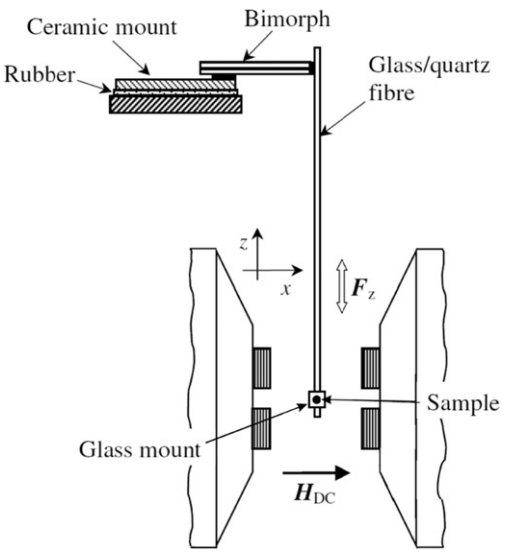

Fig. 15 Side view of a vertical gradient AGFM setup. A vertical ac field gradient is generated by means of a suitable coil arrangement. The vibration of the sample (magnetic moment pointing along the *x*-axis) along the *z*-axis is transmitted to the cantilevered piezoelectric bimorph using a glass or quartz fibre mounted to the tip of the bimorph. Adapted from Fiorillo, F., 2010. "Measurements of magnetic materials". Metrologia 47, S114–142.

proportional to the Lorentz velocity $v_{Lor} = -m \times E$, generates an additional magnetic contribution to the amplitude of the reflected light: the secondary Faraday amplitude F for transmission and the secondary Kerr amplitude K for reflection. The superposition of N with F or K then leads to magnetisation-dependent polarisation rotations.

Depending on the relative orientation of magnetisation, plane of incident light, and polarisation direction of illuminating light, three geometries can be defined: polar, longitudinal, and transverse (Hubert and Schäfer, 1998). In case of the polar effect, which is depicted in **Fig. 16(a)**, the magnetisation is perpendicular to the sample surface. If the magnetisation lies both in-plane, i.e., parallel to the sample surface, and along the plane of incidence, the longitudinal effects are obtained as shown in **Fig. 16(b)**. For transverse orientation, illustrated in **Fig. 16(c)**, the magnetisation is in-plane and perpendicular to the plane of incidence, which only leads to a magneto-optical effect in reflection.

The measuring principle of a typical MOKE magnetometer is the following. A laser, e.g., a He–Ne model with $\lambda = 633$ nm, or monochromatic light emitted by a high-intensity diode is passed through a polariser (generally a Glan–Thompson prism) set to a predefined polarisation. The light then passes through an objective lens and is focused onto the region of interest of the sample. Due to the interaction between the polarised light and the domain structure of the sample, the polarisation plane of the reflected signal is rotated, with the magnitude of the rotation being proportional to the local magnetisation. Next, the signal passes through a birefringent photoelastic modulator (PEM) to improve the signal-to-noise ratio. In addition, the PEM also introduces a phase-retardation (e.g., $\lambda/4$) allowing to separate Kerr rotation and ellipticity components of the MOKE signal. The modulated signal is used as a reference signal for a lock-in amplifier. The beam is finally transmitted through an analyser and focussed by a second objective lens onto a photosensitive diode, which provides the input signal for the lock-in amplifier. By plotting the relative Kerr intensity as a function of the applied magnetic field, the magnetisation process of thin films is characterised.

The advantages of MOKE magnetometry include high temporal and spatial resolution, low cost, as well as the possibility to perform optical measurements during preparation or treatment of a sample inside a vacuum chamber. In contrast to the other techniques discussed in this section, MOKE is a local probe as the magnetisation is only probed at the position of the laser, offering the possibility of spacial magnetisation mapping by scanning over the sample surface.

4 Soft and Hard Magnetic Materials

Based on their specific hysteretic behaviour, ferromagnetic materials can be classified into two distinct categories, soft magnetic and hard magnetic materials, which we will discuss in the present section.

4.1 Soft Magnetic Materials

Soft magnetic materials exhibit low coercivities (typically < 1 kAm^{-1}), high permeabilities, low remanence. and narrow hysteresis loops (see **Fig. 17**). They can be easily magnetised by low magnetic fields owing to their high permeabilities, but only retain a fraction of their magnetisation when the magnetic field is switched off. As a result of their low coercivities, they can be demagnetised by applying small magnetic reversal fields. When designing a soft magnetic material for a particular application, a trade-off

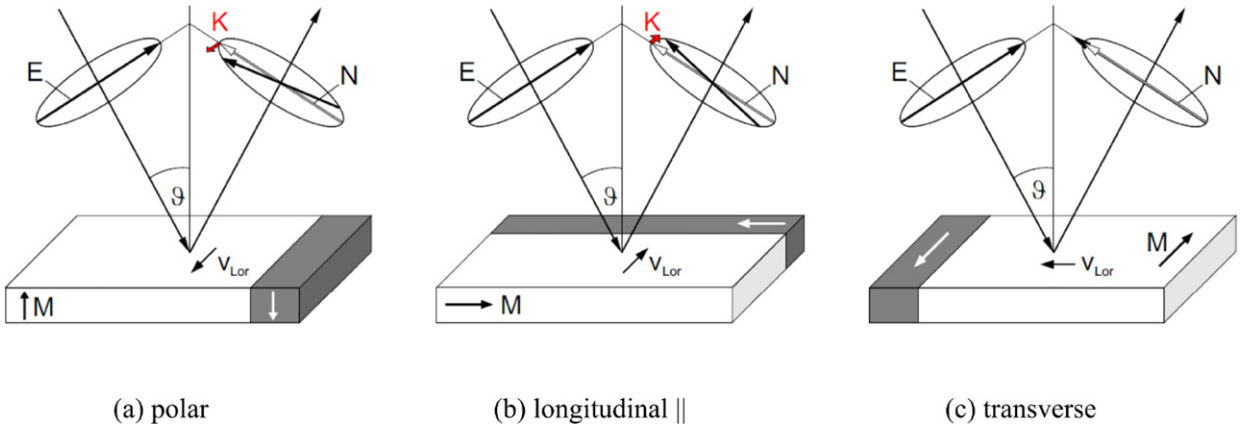

Fig. 16 Schematic representation of the different geometries of the magneto-optical Kerr effect. Adapted from Hubert, A., Schäfer, R., 1998. Magnetic Domains: The Analysis of Magneticmicrostructures, Springer.

between high saturation magnetisation, maximum permeability, minimum coercivity, low magnetostriction, low ac losses, low temperature dependence and a high thermal stability of its magnetic properties, and cost has to be made. While some of these properties are strongly sensitive to grain size and trace impurities, others are influenced by the material structure. For example, the addition of trace amounts of C, N, or S is known to increase hysteretic losses, whereas alloying with Fe, Co, Ni has to opposite effect (Cardarelli, 2000).

Soft magnetic materials include Fe in various degrees of purity (e.g., Armco and carbonyl iron), low-carbon steels and silicon steels, Fe–Al–Si alloys (e.g., Alfer, Alperm, and Sendust), Fe–B alloys (e.g., Finmet and Metglas), Fe–Co alloys (e.g., Hyperco and Permendur), Fe-Cr alloys (e.g., Ferrochrome), Fe–Ni alloys (e.g., Mumetal, Permalloy, Supermalloy, Invar, Perminvar, and Hypernik), spinel ferrites (e.g., Ni–Zn, Mn–Fe, and Ni–Co), as well as synthetic garnets (e.g., YIG) (Cardarelli, 2000). In **Fig. 18**, the ranges of coercivity and saturation magnetisation of selected soft magnetic materials are shown.

4.1.1 Iron

Pure iron has an excellent soft magnetic properties, but suffers from the fact that even small amounts of impurities cause them to significantly deteriorate. In addition, it has a high conductivity which leads to large eddy-current losses and practically prohibits the use of pure iron for ac applications. These problems have been solved by progress in steel metallurgy over the years, which have lead to a steady increase in purity of low-carbon steels made from powdered iron carbonyl (Tumanski, 2011). Despite the deteriorated magnetic properties due to the presence of C, its low price and good mechanical properties make low-carbon electrical steel an acceptable material for many cheap and basic applications and devices.

4.1.2 Silicon steels

Drastically better electrical performance as well as improved mechanical and magnetic properties were achieved by the invention of silicon steel in 1900. Depending on the exact amount of Si in the composition, either nonoriented, isotropic alloys or grain-oriented, anisotropic alloys, which are typically rolled into thin sheets, can be produced. Today, these two types of materials account for roughly 40% and 25%, respectively, of the global market for soft magnetic materials (Coey, 2009).

4.1.3 Ni–Fe alloys

Ni–Fe alloys provide the most versatile system of soft magnetic materials available today. Depending on the nickel content, which varies in the range from 30 to 80 wt%, either permeability, saturation magnetisation, or electrical resistance are maximised for high, medium, or low Ni concentrations, respectively. Both anisotropy and magnetoelasticity constants are almost zero around a nickel concentration of 80%. Though it is not possible to obtain zero value of both parameters simultaneously in binary Ni–Fe alloys, the addition of small amounts of Mo and Cu makes it possible to obtain a nonmagnetostrictive material with zero anisotropy (Warlimont, 1997). This has permitted to design alloys which exhibit ideal soft magnetic properties called Permalloys in order to indicate that high permeability is their outstanding feature. Annealing treatments combined with field-induced uniaxial anisotropy are used to control and adjust the magnetisation process to specific applications.

4.1.4 Co–Fe alloys

Co–Fe alloys are significantly less versatile and much more expensive due to the costly cobalt. Crystalline anisotropy and magnetostriction have minima at different Co contents, allowing compositions to be optimised for different applications. For example, Permendur, $Fe_{50}Co_{50}$, has an unrivaled saturation magnetisation of 2.45 T which is the highest value of all magnetic materials at room temperature, whereas NiFeCo alloys called Perminvar exhibit constant permeability across a wide field range.

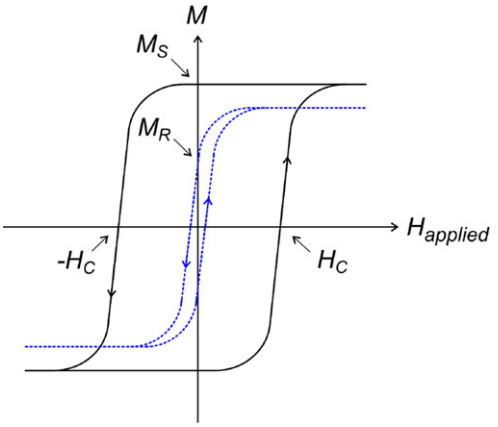

Fig. 17 Schematic hysteresis loops of a soft magnetic (dashed curve) and a hard magnetic (solid curve) material.

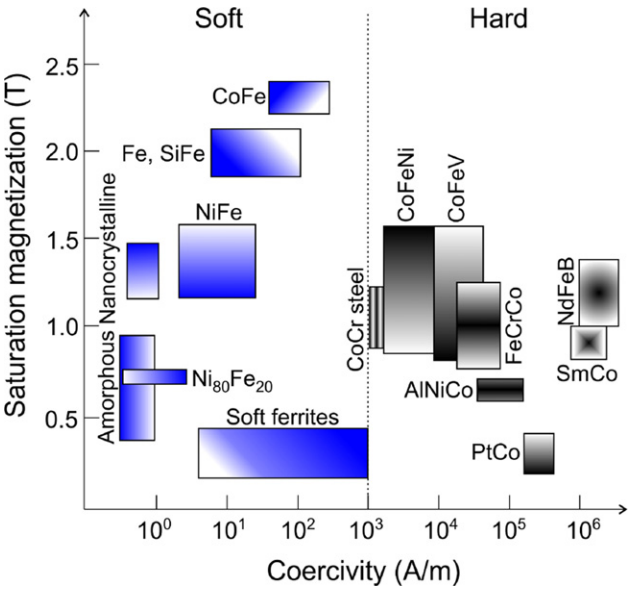

Fig. 18 Ranges of coercivity and saturation magnetisation of commercially available magnetic materials as an example of products offered by Vacuumschmelze. Reproduced from Tumanski, S., 2011. Handbook of Magnetic Measurements. Boca Raton: CRC Press.

Adding 2% vanadium improves machinability, delays the structural ordering transformation, and drastically increases electrical resistivity, thus reducing eddy current losses, without sacrificing the magnetic properties.

4.1.5 Amorphous alloys

Amorphous alloys, also known as metallic glasses, are compositions close or equal to the form $T_{80}M_{20}$, where T = Co, Fe, Ni and M = B, Si. There are two major groups of amorphous alloys: Fe-rich glasses of FeSiB-type with high saturation magnetisation and Curie temperature, and Co-rich alloys of CoFeMnSiB type with zero magnetostriction and very high permeability. Metallic glasses are produced by melt-spinning thin tapes or ribbons which together with their very high resistivity allows them to compete with ferrites in the medium frequency range up to 100 kHz. Fully or partially crystalline materials, e.g., Finmet, obtained by annealing amorphous alloys, exhibit excellent soft magnetic properties that can easily compete with the best permalloys.

4.1.6 Soft ferrites

Soft ferrites are a very important group of soft magnetic materials thanks to their low raw material and production cost as well as ease of fabrication into a wide variety of shapes. Due to their ferrimagnetic nature, saturation magnetisation and permeability are mediocre compared to metallic ferromagnets. However, their insulating character makes them suitable for high-frequency applications since eddy-current losses are strongly suppressed. Structurally, ferrites are iron spinels, and the two major types of

high permeability ferrites are Mn-Zn and Ni-Zn ferrites, which can be operated at frequencies up to around 1 MHz and from 1 to 300 MHz or even more, respectively (Coey, 2009). The latter have a lower saturation magnetisation, but much higher resistivity.

4.1.7 Rare-earth iron garnets

Another group of ferrites are iron garnets containing trivalent rare earth ions such as, e.g., yttrium iron garnet (YIG). This is a perfect insulating crystal with minimal anisotropy as well as very narrow ferromagnetic resonance linewidth, making it an excellent material for applications in the GHz range.

4.2 Hard Magnetic Materials

Hard magnetic materials have high coercivities (typically >10 kAm^{-1}), low permeabilities, high remanence, high maximum energy products, and wide hysteresis loops (see **Fig. 17**). Owing to their ability to retain a high residual magnetisation even after a high magnetic field has been switched off, they are also referred to as permanent magnets. The ranges of coercivity and saturation magnetisation of selected hard magnetic materials are depicted in **Fig. 18**. From both a historic and commercial point of view, hard magnetic materials can be classified into five major categories (Cardarelli, 2000), whose general properties and preparation methods will briefly be described in the following.

4.2.1 Martensitic steels

Magnetic steels were originally made by rapid quenching alloys of iron with up to 2 wt% C, known as high-carbon steel, at temperatures around 800°C. The addition of carbon reduces the saturation of iron, but coercivity and energy product are vastly improved by means of internal stresses and inhomogeneities connected with a lattice transformation. The resulting steels were mechanically very hard, which is where the term hard magnetic originates from. In the early 20th century, additional alloying elements such as tungsten and later chromium were added in order to further increase the coercivity to values between 3 and 6 kAm^{-1}, eventually leading to the discovery of stainless steel. In 1917, cobalt was introduced as an alloying element by the Japanese which allowed to reach coercivities of up to 25 kAm^{-1} and energy products of up to 8 kJm^{-3} (Cardarelli, 2000). Cobalt steels including the addition of Mo or V, known as Remalloys or Vicalloys, respectively, had commercial use at that time, but quickly became obsolete with the discovery of Alnico in the 1930s.

4.2.2 Precipitation-hardened alloys

In the early 1930s, the Japanese experimented with carbon-free Fe alloys containing aluminum, nickel, and later also cobalt, which lead to significant improvements over the magnet steels used earlier. Alnico, an acronym for aluminum–nickel–cobalt, combined high remanence of up to 65 kAm^{-1}, energy products of up to 80 kJm^{-3}, low temperature expansion coefficient, high working temperature of 550°C, and reasonable cost, making it the material of choice at that time (Cardarelli, 2000). The improved magnetic properties of these alloys are the result of a precipitation phase introduced through tempering and annealing in high magnetic fields performed prior to the final hardening treatment. Shape anisotropy, determined by size, shape, and orientation of the nanoparticles formed during heat treatment, supplemented in many cases by an increase in crystalline anisotropy, is responsible for their high coercivity. Changing the production process of Alnico magnets from casting to sintering not only reduces their surface roughness as well as their considerably high hardness and brittleness, but also makes it possible to produce very small magnets, even with complicated shapes.

4.2.3 Superstructure alloys

In the two-component alloys platinum–cobalt, platinum–iron, and platinum nickel, hardening occurs by a similar precipitation mechanism, but with an additional superstructure formation, i.e., the precipitated phase is ordered, while the underlying matrix is unordered. A Pt–Co alloy containing approximately equal numbers of atoms of each element, attains a coercivity of ~ 500 kAm^{-1}, which is only surpassed by cobalt-rare earth magnets, as well as an energy product of ~ 75 kJm^{-3}, a value only exceeded by the best Alnico magnets available at the time of its discovery in the 1950s (Goldmann, 1999). These alloys are characterised by excellent mechanical and corrosion resistant properties. Though their magnetic properties are superior to most other permanent magnet materials with the exception of Nd–Fe–B, they were replaced by this very same material because of the expensive price of platinum.

4.2.4 Ceramic hard ferrites

Hard hexagonal ferrites such $BaO \cdot 6Fe_2O_3$, $SrO \cdot 6Fe_2O_3$, or $Pb \cdot 6Fe_2O_3$, were developed since the 1950s and represent the most widely used permanent magnet material today due to their extremely low costs and ease of production. These magnets are made by a multi-step powder metallurgy process, which in the absence of a magnetic field produces isotropic magnets, whereas the presence of a strong magnetic field aligns the easy c axis to produce anisotropic magnets with almost doubled remanence, but unchanged coercivity. In addition, the final heat treatment can be managed such that coercivity is raised at the expense of remanence, and vice-versa, leading to remanence- or coercivity-biased anisotropic products (Heck, 1974). While their manufacturing process allows them to be molded into many complex shapes, the resulting ceramic magnets are inherently brittle and frequently contain imperfections. The coercivity of hard ferrites can be as high as 250 kAm^{-1}, but due to their low remanence, the corresponding energy product is only of the order of 40 kJm^{-3} (Coey, 2009).

4.2.5 Rare-earth element alloys of SmCo- and NdFeB-type

The next major milestone in the technology of permanent magnetic materials was the development of alloys combining rare-earths and the transition metals Ni, Fe, and Co in the late 1960s. The two best known and most important compositions of this family are $SmCo_5$ and Sm_2Co_{17}. The 2–17 material has superior magnetic properties, but contains more of the costly samarium and is difficult to produce consistently, which is why only the 1–5 material is of commercial significance (Goldmann, 1999). With values of up to 160 kJm^{-3} for $SmCo_5$ and 240 kJm^{-3} for a Sm_2Co_{17}-based alloy, respectively, they held the record for the highest energy product for many years (Cardarelli, 2000). Both SmCo-types exhibit coercivities in excess of 1000 kAm^{-1}, and have excellent corrosion resistance, thermal stability up to 300 °C, as well as high resistance to demagnetising influences. Their fabrication process involves powder metallurgy, with sintered magnets having superior magnetic properties over their bonded counterparts.

The most recent development and yet another significant step up from SmCo magnets was the discovery of $Nd_2Fe_{14}B$ magnets. It was made both simultaneously and independently by Sumitomo Special Metals in Japan and General Motors in the USA in 1984, and mainly driven by the limited availability and volatile price fluctuations of Co which were problems for the SmCo magnet market (Goldmann, 1999). The Japanese production method was based on a powder metallurgy process, which involved casting an ingot, breaking it up into powder, pressing, sintering, and heat treating. The resulting NdFeB magnets have energy products of up to 450 kJm^{-3}, making them the most powerful commercialised permanent magnets available today (Coey, 2009). High-coercivity grades of NdFeB are much more prone to oxidation than any other magnet alloy due to the highly reactive nature of their Nd-rich phase, which is why usually a protective nickel coating is applied in the final production steps if the material is to be exposed to humidity or chemically aggressive substances. In the USA, melt-spinning was used to produce fine wires or ribbons of alloy, followed by powdering and pressing. While this process was much simpler, the magnetic and microstructural properties obtained by it are less good as they are highly sensitive to the quench rate of the ribbons. As NdFeB magnets posses excellent magnetic properties, come in a wide range of grades, sizes, and shapes, are relatively inexpensive, and their raw materials are available in abundance, they can be used to replace traditional permanent magnets such as Alnicos or even SmCo by higher performing, more efficient, more compact, and more cost-effective designs and devices. Despite their high energy product, NdFeB magnets only have a relatively low Curie temperature which limits their maximum application temperature to 200 °C. However, their temperature characteristics can be improved by the addition of Co and Dy at the expense of increased production costs.

5 Applications and Practical Aspects of Hysteresis

As discussed in the previous section, soft and hard magnetic materials posses vastly different magnetic properties which are exploited in a large number of applications that fall into two categories: ac and dc. In this section, we will give a general overview of their main applications classified by their operating frequency range (Coey, 2009).

5.1 Applications of Soft Magnetic Materials

For dc or static applications (<1Hz) such as in electromagnets, relays, and shielding, soft iron, Co–Fe (Permendur), and Ni–Fe (Permalloy) are the preferred materials. In the low frequency range from 1 Hz to 1 kHz, silicon steel, Permalloy, Finmet, and magnetic glasses are the materials of choice for use in transformers, generators, and motors. For audio frequencies (100 Hz to 100 kHz), Permalloy foils, Finmet, amorphous alloys, Fe–Si–Al powder (Sendust) as well as Mn–Zn ferrite are used for inductors, transformers in switched mode power supplies, and TV flyback transformers. The radio frequency range between 0.1 and 1 GHz exploits Mn–Zn and Ni–Zn ferrites for inductors and antenna rods. Lastly, in the microwave range beyond 1 GHz, YIG and Li ferrite are the materials of choice for use in microwave isolators, circulators, phase shifters, and filters.

5.1.1 Static applications

DC applications of soft magnetic materials such as in electromagnets require optimum flux guidance and concentration, which is why materials such as pure soft iron or $Fe_{50}Co_{50}$ combining highest saturation magnetisation with both very low remanence and coercivity are used. Passive magnetic shielding prevents an external weak dc or low-frequency ac magnetic field from penetrating into an internal space, where it could interfer or even prevent proper device or circuitry operation. Shielding materials either deflect magnetic flux by guiding it around the sensitive volume to be protected, or contain magnetic flux in direct vicinity of the device generating it. DC shields are typically made of Permalloy or Mumetal, which are both insensitive to shock and strain because of their vanishing anisotropy and magnetostriction. Formable and flexible shielding can be made from Finmet sheets as well as woven Co-rich metallic glass ribbons (Coey, 2009).

5.1.2 Low frequency applications

Low frequency applications consume the largest amount of magnetic material produced and typically involve power generation and distribution at mains frequencies of 50 or 60 Hz as well as of 400 or 800 Hz in the case of military and avionics applications, respectively. Soft iron cores are generally used to generate and guide flux in inductors and generators. Eddy-current losses are reduced by employing thin laminations of electrical steel, such as in the case of induction motors or high-resistivity materials. Isotropic sheets of silicon steel are suited for generators and motors, where the flux direction varies continually during operation.

In transformers, where the flux axis is fixed, it is advantageous to use grain-oriented sheet steel having an easy axis to reduce losses. Magnetic amplifiers and fluxgate magnetometers rely on saturable soft cores and are typically made from textured Permalloy. Transducers, linear actuators, and sonar make use of a Tb-Dy-Fe alloy named Terfenol, which possess the highest magnetostriction of all known alloys (Coey, 2009). Cost is still the main consideration for almost all of these applications, but higher operating efficiency and thus lower energy consumption become more and more important.

5.1.3 High-frequency applications

High frequency applications rely to a very large degree on the use of soft ferrites, which are used in chokes, inductors such as loading coils, wideband pulse transformers, narrowband resonant filters, antennae, high-frequency transformers for switch-mode power supplies, and broadband amplifiers.

At very high frequencies, such as the in microwave range from 300 MHz to 300 GHz, electrical energy is transmitted by means of electromagnetic waves travelling in waveguides instead of currents flowing in wires and through circuits. Devices operating in this frequency range are generally based on YIG and exploit the nonreciprocal interaction between a ferromagnetic medium and an electromagnetic field by either resonantly absorbing microwaves or producing the microwave Faraday effect as a result of circular birefringence far from resonance. The differences between the permeabilities of oppositely rotating, circularly polarised microwaves are exploited in different ways (Coey, 2009). Resonant isolators absorb microwaves propagating in one direction, but not the other, when the ferrite material is magnetised to its resonance condition. Phase shifters exploit the very same difference, but above and below resonance. Circulators with either three or four ports transmit signals from one port to the next one, while the other ports remain isolated. Faraday rotators change the polarisation plane of microwaves.

5.2 Applications of Hard Magnetic Materials

In general, the principal applications of hard magnetic materials are magnetic recording media and permanent magnets. They are classified according to the nature of their magnetic flux distribution which can either be spatially uniform or nonuniform, as well as static or time-varying (Coey, 2009). The physical effects exploited in their applications include force, torque, induced electromotive force, magnetoresistance, or Zeeman splitting.

5.2.1 Static applications in uniform magnetic fields

Though the field generated by a magnetic dipole is nonuniform and anisotropic, it is possible to achieve a uniform field by arranging pieces of magnetic material magnetised along different directions such that the sum of their individual contributions results in a uniform field over a limited volume in space. For example, in a cylindrical magnet with a hollow bore, a uniform field can be generated in the central region. An open square cylinder or a design with flat cuboid magnets surrounded by a soft iron yoke is used in nuclear magnet resonance and magnetic resonance imaging equipment, accelerator magnets, and magnetic refrigeration devices. Halbach cylinders, made from wedge shaped magnets whose magnetisation varies around a common axis, produce fields of the order of 1 T used for magnetic annealing to set the exchange bias in spintronic devices.

5.2.2 Static applications in non-uniform magnetic fields

Non-uniform, transverse multipole fields and uniform field gradients can be generated by modified cylindrical magnet structures. For example, external multipole magnets, which are variants of a normal Halbach cylinder and produce a multipole field outside the cylinder while the field in their interior is zero, are used as rotors of electric motors based on permanent magnets. The pattern of an unrolled Halbach cylinder gives a single-sided magnet with a strong field on one side, but none on the other side, which is used for fridge magnets and other holding magnets based on bonded ferrites.

Highly brilliant, polarised synchrotron radiation is generated by passing a high-energy electron beam through insertion devices such as wigglers and undulators, whose sinusoidal transverse field of wavelength λ stimulates the electrons to radiate with a frequency c/λ Klystrons and magnetrons employ permanent magnets to generate microwave radiation from electrons.

Magnetic separators use rare-earth permanent magnets to generate complex flux patterns with huge spatial fluctuations (10^2 to 10^5 Tm^{-1}) for separating ferrous and nonferrous materials based on their different magnetic susceptibilities. There are three basic configurations: pulleys, drums, and suspended magnets, which are used in either high-gradient or open-gradient magnetic separators. Magnetic separation is used routinely in, e.g., the mining industry for the beneficiation of iron ores such as magnetite from slag and slurries, the recycling and scrap processing industries for the recovery of potentially salable iron and steel from municipal solid waste, and water purification systems for eliminating heavy metals from waste water.

5.2.3 Dynamic applications

Permanent magnet-based, variable flux sources such as a double Halbach-cylinder, a four-rod mangle, or an external Halbach-cylinder with a soft iron sheath are compact and do not require cooling, making them ideally suited for compact benchtop measurement systems such as VSMs.

When linear and rotary motion need to be combined, but no mechanical contact point is allowed, gears or bearings based on axially magnetised permanent magnets can be used that are well suited to high-speed rotary suspensions in turbopumps or flywheels used for energy-storage.

Magnetic sensors detect a varying magnetic field in an airgap using Hall effect or magnetoresistance probes which deliver a field dependent voltage from which, e.g., speed and position can be determined in a noncontact fashion.

Actuators, electromechanical devices having a limited linear or angular displacement range, use permanent magnets in three different configurations. Moving coil actuators are used in loudspeakers as well as ear- and headphones, with low-end designs employing flat ferrite ring magnets, mid-range designs using cylindrical Alnico or rare-earth magnets, and high-end models utilizing $Nd_2Fe_{14}B$ magnets. Voice-coil and flat coil actuators are used for positioning of read and write heads in computer hard-disc drives, requiring Nd–Fe–B magnets with the highest energy product available. Moving-iron actuators are designs that are used for print hammers in dot-matrix printers as well as in reed switches.

Motor designs have evolved remarkably since the availability of ferrite and rare-earth magnets as well as polymer-bonded magnets in any desired shape. Small dc motors are typically found in domestic appliances such as washing machines and consumer electronics such a compact disc drives. DC servo motors power tools and robots, whereas stepper motors are used for precision control clocks, watches, as well as linear motion and rotation stages.

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